

Your company has started an initiative to upgrade its existing video streaming service for its customers. The company needs to ensure that its SLA for the service to its customers is met. You have an SLA of 99% defined for service availability. What should be the value of your SLO?

- A) 99.5%

B) 89%

C) 98.99%

D) 99%
- ✔

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✖

Explanation

You should use the value of 99.5% for the service level objective (SLO). This is because you need to keep an SLO value that is higher than the service level agreement (SLA), which helps you get early warnings when there is a danger of not meeting the SLA. Failure to meet a set SLA can incur financial penalties for the company. SLAs are visible to customers that use a service while SLOs and SLIs are internal to the development and operations team. If you do not meet an SLO, then there is a possibility of not meeting the SLA, which can cause issues for the company.

The SLO of 99% is incorrect because this value does not allow you to have an early warning of SLAs not being met. You need to keep the SLO value higher than the SLA, as a rule. The SLO of 99% is the same as the SLA of 99%, which indicates that failure to meet the SLO will mean failure to meet the SLA and indicates that there is no early warning possible for the SLA not being met. You need to specify an SLO that exceeds the SLA. An SLO specifies a definite target for a service, which can be 99.999% uptime. SLOs are defined based on user experience and can address user-centric needs, such as latency.

The SLO of 89% and 98.99% are both incorrect because these SLO values are lower than the SLA value of 99%. This raises an unacceptable situation where SLOs will not meet the SLA. For this reason, you need to ensure that the SLO value exceeds the SLA to create a system where there is no possibility of the SLA not being met.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Explain how IT operations need to adapt to thrive in the cloud

References:
[Services.google.com > Increasing Business Value with Better IT Operations: A guide to Site Reliability Engineering \(SRE\)](#)

Related Material

You need to store the events generated by various Google Cloud services in buckets that need to be localized to specific regions for regulatory and compliance requirements. Which Google Cloud service would be the most appropriate choice for this?

- A) Cloud Endpoints

B) Cloud Logging

C) Cloud Trace

D) Cloud Profiler
- ✖

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Explanation

You would use Cloud Logging for this scenario. Cloud Logging is a log management service that is fully managed. It allows you to store, search, and analyze exabytes of log data as well as set up alerts and notifications. Cloud Logging has Logs Explorer, which lets you use flexible queries for searching, sorting, and analyzing logs. You can also utilize log buckets to store and process log data based on regional compliance requirements. Cloud Logging also has an Error Reporting feature that performs a smart collection of errors in logs into error groups. Cloud Logging also allows your security teams to keep audit trails using its Cloud Audit Logs feature, which provides greater transparency in activities related to administration. Cloud Logging uses the Cloud Logging Application Programming Interface (API) for collecting log entries, which are passed through a Logs Router. The Logs Router uses existing filters to identify the log entries meant for exclusion, ingestion, or import.

You would not use Cloud Profiler for this scenario. Cloud Profiler is used to find and remove any possible performance issues from your applications currently in production. This is accomplished by Cloud Profiler regularly profiling the consumption of resources across your production applications. Cloud Profiler creates an analysis of the performance of an application by collecting data related to CPU usage and memory allocation.

You would not use Cloud Trace for this scenario. Cloud Trace is a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights into its performance.

You would not use Cloud Endpoints for this scenario. Cloud Endpoints provide a system for managing APIs that enables monitoring, analyzing, and securing APIs as well as specifying quotas for them. You can monitor the key operational metrics for the APIs using Google Cloud Console and gain usage insights with BigQuery, Cloud Trace, and Cloud Logging.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Identify Google Cloud solutions for cloud resource monitoring and application performance management

References:
[Cloud.google.com > Cloud Logging](#)

Related Material

Your operations and development teams need to create a plan for ensuring service availability while adding some key enhancements. For this purpose, which of these can you consider valid service level indicators (SLIs) to meet your service level objectives (SLOs)? (Choose two.)

- A) Number of successful login attempts

B) Percentage of correct user responses

C) Amount of memory consumed by the service

D) CPU utilization for the service

E) Number of API calls generated for the service
- ✔

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Explanation

The percentage of correct user responses and the number of successful login attempts are both valid service level indicators (SLIs) that you can use to meet your service level objectives (SLOs). SLIs measure the quantity of a specific aspect of the service that is relevant to the user. SLOs are defined based on user experience and can address user-centric needs, such as latency. SLIs help gauge if SLOs are met. A combination of SLOs and SLIs work towards meeting service level agreements (SLAs), which specify the exact agreement made with the users of the service.

CPU utilization for the service, the amount of memory consumed by the service, and the number of application programming interface (API) calls generated for the service are all not the best options for SLIs as these are internal to the system and not user-centric. By selecting SLIs that capture performance aspects that are relevant to the user, you can better meet SLA needs.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Explain how IT operations need to adapt to thrive in the cloud

References:
[Services.google.com > Increasing Business Value with Better IT Operations: A guide to Site Reliability Engineering \(SRE\)](#)

Related Material

Question #4 of 60

Question ID: 1495283

You work for a software development company that makes two deployments a day for its workplace collaboration tool, which is heavily in use and growing daily. Your development team has come up with a lot of new features that it wants to release, but the specified error budget for the month has been consumed. How will you roll out future deployments?

- A) Once a day

✖
- B) Once a month

✖
- C) Halt deployments

✔
- D) Once a week

✖

Explanation

You will halt deployments. This is because when new features are added to a service, new bugs may be introduced, which can eat into an existing error budget. There can then come a stage where the error budget for a given period is exhausted, and management will need to halt development of new features until the error budget is restored. For example, when a company that deploys code changes twice a day consumes half its error budget, it needs to reduce code deployments to once a day. When three-fourths of the error budget is consumed, the deployments can be reduced to once a month. When the error budget hits zero for a month, deployments will need to be halted.

The other options are incorrect because you cannot roll out any new deployments when the error budget is consumed for the month. This can cause a failure to meet the service level agreement (SLA) with the service users and cause financial loss for the company. The error budget is the measure of the system's possible downtime, which is the difference between the service level objective (SLO) and the total availability figure of 100%. When an SLO is agreed upon, its corresponding service downtime or error budget is also agreed upon. An SLO specifies a definite target for a service, which can be 99.999% uptime. SLOs are defined based on user experience and can address user-centric needs, such as latency.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Explain how IT operations need to adapt to thrive in the cloud

References:
[Services.google.com > Increasing Business Value with Better IT Operations: A guide to Site Reliability Engineering \(SRE\)](#)

Related Material

Question #5 of 60

Question ID: 1495226

Your company has a two-tier application with a web-based frontend for interacting with users and an RDBMS backend that stores user data. You need to find a way to gradually move your systems from on-premises to a cloud deployment. How should you implement a tiered hybrid cloud deployment?

- A) Migrate backend applications to a public cloud

✖
- B) Deploy frontend applications to a public cloud

✔
- C) Store data that is restricted by regulations on a public cloud

✖
- D) Use a private cloud for performing changes to frontends

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Explanation

You should deploy frontend applications to a public cloud. Frontend applications require frequent changes and updates as they face users and are performance centric. By moving frontend applications to a public cloud, you can have an automated continuous integration and continuous delivery (CI/CD) system for efficient deployment of updates to applications.

You would not migrate backend applications to a public cloud. Backend applications can have various system dependencies that are very complex, and the task of migrating these applications to a public cloud can be an involved and challenging process. By focusing on moving the frontend applications to the cloud first, you can perform cloud integration faster and utilize cloud-native services for increased availability and resiliency for the system.

You would not store data that is restricted by regulations on a public cloud. Any data that is subject to regulatory restrictions should ideally be stored in an on-premises environment. This approach ensures greater security of the data.

You would not use a private cloud for performing changes to frontends. This is because a public cloud system offers features like CI/CD for rapid and automated development and updates for applications. Additionally, cloud services, including Cloud Internet of Things (IoT) and Cloud Content Delivery Network (CDN), can be useful for creating user interfaces, data ingestion from IoT sources, and creating application programming interfaces (APIs).

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Hybrid and multi-cloud architecture patterns](#)

Related Material

Question #6 of 60

Question ID: 1497405

You want to ensure maximum security of your sensitive data and also leverage cloud-native services like serverless computing. Which cloud deployment should you opt for?

- A) Public Cloud

✖

B) Hybrid Cloud

C) Multi-Cloud

D) Private Cloud

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Explanation

You should opt for a hybrid cloud infrastructure model. In a hybrid cloud system, a company uses its own data center along with a public cloud system to achieve its business goals. A hybrid cloud always contains a private cloud. A hybrid cloud deployment allows a company to run its applications on-premises and use the public cloud for bursting when user demands go beyond the limits of the data center.

You would not opt for a private cloud infrastructure. In this deployment, all the infrastructure and applications that are part of the cloud or virtualization system are running on systems that are inside a company's on-premises data center. This deployment option does not include systems that are a part of a public cloud service offering.

You would not opt for a public cloud infrastructure. In this deployment, the infrastructure includes applications that are fully functional in a cloud and have no on-premises portions. This deployment option may not be suitable for situations where sensitive data needs to be stored locally and not on a public medium.

You would not opt for a multi-cloud infrastructure. In this deployment, the data is still stored on a public system, which can cause security concerns for the company. A multi-cloud deployment involves using multiple public CSPs where each public cloud provides a different service for the company. This allows a company to use one public cloud for database services and another one for analytics. This deployment option lets a company use a preferred cloud provider for specific services. Remember that using multiple public CSPs indicates a multi-cloud infrastructure.

Objective:

Infrastructure & Application Modernization

Sub-Objective:

Learn what modernizing IT infrastructure with Google Cloud means

References:

[Cloud.google.com > Hybrid and multi-cloud architecture patterns](#)

Related Material

Question #7 of 60

Question ID: 1495233

Your company develops business software for various global markets. You need to ensure that your CI/CD pipeline is secure and any unsafe images are blocked from deployment. Which Google Cloud service should you use for this?

A) BigQuery

B) Dataproc

C) Cloud Build

D) Firestore

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Explanation

You should use Cloud Build for this scenario. Cloud Build is used to build, test, and deploy systems using a continuous integration and continuous delivery (CI/CD) platform, which is serverless. It allows a company to integrate its existing legacy systems and tools into a modern build system. This is made possible using third-party applications and customized steps for builds. Cloud Build also has vulnerability scanning features for CI/CD that prevents the deployment of any images that are unsafe. Some key features of Cloud Build include:

- Fast build time using VMs with fast CPUs provisioned to cache source code and dependencies.
- Deployment automation using pipelines with integrated use of Firebase, Google Kubernetes Engine (GKE), Cloud Functions, and App Engine.
- Multicloud deployment using multilanguage support for builder images used and containerized tasks, which are portable across clouds.
- Strong privacy through Google Cloud security, which allows the use of repositories for private source code inside private networks.

You would not use BigQuery for this scenario. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence (BI), and geospatial analytics ready for use. It also has a serverless architecture that allows you to use SQL queries to gain key organizational data. BigQuery has a scalable analysis engine that has a distributed structure and allows you to perform queries on petabytes of data within minutes.

You would not use Firestore for this scenario. Firestore is a serverless document database that is scalable and fully managed. It enables fast development for applications for the internet, mobile, and the Internet of Things (IoT), which can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features. It also allows you to customize various rules for security and data validation to protect your data. Firestore can be integrated with BigQuery and Firebase. Firestore is an upgrade to the older service called Datastore, which provides a NoSQL database that is scalable.

You would not use Dataproc for this scenario. Dataproc provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, Extract, Transform, and Load (ETL) operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

Objective:

Infrastructure & Application Modernization

Sub-Objective:

Learn what modernizing IT infrastructure with Google Cloud means

References:

[Cloud.google.com > Cloud Build](#)

Related Material

Question #8 of 60

Question ID: 1495947

Your company has a photo sharing service that is heavily in use. You have created an SLO for the service. You need to find a way to check the status of your SLO and also have the error budget computed automatically. Which Google Cloud service would be suitable for this requirement?

A) Cloud Monitoring

B) BigQuery

C) Cloud Vision

D) Looker

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Explanation

Cloud Monitoring can be used to create and use service level objectives (SLOs), service level indicators (SLIs), and error budgets. You can specify an SLO for a month, and Cloud Monitoring computes the corresponding error budget automatically. It can be used to monitor the complete infrastructure of a cloud deployment. It provides a view into the uptime, health, and performance of cloud systems. For this, Cloud Monitoring gathers relevant metadata, events, and metrics from sources that include uptime probes, services, app instrumentation, alerts, notifications, and other components of applications. It allows you to stop issues from happening, and it allows you to identify trends. It also reduces overhead related to monitoring through a service that is integrated and provides metrics, dashboards, and alerts all in one place. Cloud Monitoring provides SLO monitoring and metrics collection for virtual machines (VMs) (using Ops Agent) and Kubernetes (using Managed Service for Prometheus).

You would not use BigQuery for this scenario. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence, and geospatial analytics ready for use, and has a serverless architecture that allows you to use SQL queries to gain key organizational data. BigQuery has a scalable analysis engine that has a distributed structure. This allows you to perform queries on petabytes of data within minutes.

You would not use Looker for this scenario. Looker is a platform for business intelligence that allows you to access intelligent data for making better business decisions. It also can be embedded into other applications to provide embedded analytics. Looker can be used to create alerts for proactively sending timely insights related to various business data. It also has dashboards that are interactive, which allow for analyzing data in real-time through a trusted data model.

You would not use Cloud Vision for this scenario. Cloud Vision is used by developers to understand image content through machine learning (ML) models accessed through an application programming interface (API). It provides image classification into various categories, object and face detection inside images, and identification of printed words inside of images.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Identify Google Cloud solutions for cloud resource monitoring and application performance management

References:
[Cloud.google.com > Cloud Monitoring](#)

Related Material

Question #9 of 60

Question ID: 1495292

Your company has various systems and services running on Google Cloud. Each of these generates events. You need a way to create alerts whenever a specific error event on the system is generated. Which Google Cloud service should you use to accomplish this?

- A) Cloud Logging**
- B) Cloud Spanner**
- C) Cloud Armor**
- D) Cloud Vision**
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Explanation

You should use Cloud Logging for this scenario. Cloud Logging is a log management service that is fully managed. It allows you to store, search, and analyze exabytes of log data, as well as set up alerts and notifications. Cloud Logging has Logs Explorer, which lets you use flexible queries for searching, sorting, and analyzing logs. You can also utilize log buckets to store and process log data based on regional compliance requirements. Cloud Logging also has an Error Reporting feature that performs a smart collection of errors in logs into error groups. Cloud Logging also allows your security teams to keep audit trails using its Cloud Audit Logs feature, which provides greater transparency in activities related to administration. Cloud Logging uses the Cloud Logging Application Programming Interface (API) to collect log entries passed through a Logs Router. The Logs Router uses existing filters to identify the log entries meant for exclusion, ingestion, or importing.

You would not use Cloud Spanner for this scenario. Cloud Spanner is a relational database service that is fully managed and ideal for critical business applications. You would use Cloud Spanner for the provision of a highly scalable database for online transaction processing (OLTP). This database will have high levels of availability and consistency worldwide.

You would not use Cloud Vision for this scenario. Cloud Vision is used by developers to understand image content through machine learning (ML) models accessed through an API. It provides image classification into various categories, object and face detection inside images, and identification of printed words inside of images.

You would not use Cloud Armor for this scenario. Cloud Armor is a service that protects websites and software applications from threats like web attacks and distributed denial of service (DDoS) attacks. It offers adaptive protection by detecting and mitigating large numbers of Layer 7 DDoS incidents using a locally trained machine learning (ML) system.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Identify Google Cloud solutions for cloud resource monitoring and application performance management

References:
[Cloud.google.com > Cloud Logging](#)

Related Material

Question #10 of 60

Question ID: 1497152

Your company has a large catalog with many stock keeping units (SKUs). The company wants to utilize efficient third-party tools for managing costs for projects programmatically in order to have greater clarity into cloud costs and their management. Which of these options will help meet this requirement?

- A) Using billing APIs**
- B) Setting resource quotas**
- C) Billing the on-premises data center**
- D) Billing the cloud service provider**
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Explanation

Using billing application programming interfaces (APIs) will help meet this requirement. With billing APIs, you can use various tools created by other vendors to manage project bills programmatically and obtain data on all stock keeping units (SKUs) in their repository and related metadata. Other methods of ensuring clarity for financial governance include using intelligence recommendations for performing rightsizing of computing resources. This helps ensure that costs are minimized by ensuring that virtual machine (VM) usage is optimized. Also, you can use billing reports from the Google Cloud Platform (GCP) console to plan, track, and ensure optimal use of resources within your cloud deployment. This is done through views of trends for current costs and forecasts for future spending. You can also export this data to Data Studio and BigQuery to create customized dashboards that provide more details on cloud utilization and its associated costs.

You would not bill the on-premises data center in this scenario. This is not a way to provide programmatic management of project billing for a cloud deployment. For programmatic management of costs and billing, you need to utilize APIs, which can provide an interface between your systems and the services offered by third parties. The on-premises data center's operations and cost management are unrelated to the cloud deployment.

You would not set resource quotas in this scenario. Setting resource quotas does not provide a way for programmatically managing billing for projects. Setting resource quotas and performing throttling needs to be done to ensure that cost overruns do not occur due to resource utilization that exceeds the specified budgets.

You will not bill the cloud service provider (CSP). You, as a customer, are billed by the CSP based on the resource consumption and service utilization of your cloud deployment. CSPs use the Pay-As-You-Go (PAYG) model of payment, where a cloud consumer only needs to pay for resources consumed and not for unused resources. For example, if a computing instance is in a terminated state, then it will not contribute to the total billable costs.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > A guide to financial governance in the cloud](#)

Related Material

Question #11 of 60

Question ID: 1497146

Your company provides recruitment consultant services for a wide user base and needs a technology solution for classifying and extracting structured data from inside user application forms. Which of these technologies would you use for this requirement?

- A) Visual Inspection AI

✖
- B) Cloud Translation

✖
- C) Document AI

✔
- D) Media Translation API

✖

Explanation

Document Artificial Intelligence (AI) is the appropriate choice for this scenario. You would use Document AI for classification and extraction of structured data from existing documents. This can enable you to find key insights and perform automation of organizational practices. You can use Human-in-the-Loop AI for workflow systems and a user interface to verify the information that is extracted from documents using Document AI.

Media Translation application programming interface (API) is not the appropriate choice for this scenario. You would use Media Translation API for the automatic translation of audio between languages. This can also be done in real-time, if required. You can use Media Translation API for programmatic translation of audio inside of your applications. Media Translation API is based on the open source Google Remote Procedure Call (gRPC) framework.

Visual Inspection AI is not the appropriate choice for this scenario. You would use Visual Inspection AI for training and deploying AI models for automatic detection, classification, and localization of any anomalies that are identified in images. This helps increase the quality of production in various industries and enables the development of advanced analytics.

Cloud Translation is not the appropriate choice for this scenario. Cloud Translation or Translation AI performs the translation of text between languages using Representational State Transfer (RESTful) APIs. You can use Cloud Translation to translate text from English to Spanish to any other language or vice versa. By using Cloud Translation API, you can provide automatic text translation features for an app or Website.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Document AI](#)

Related Material

Question #12 of 60

Question ID: 1495273

You work for a government agency that has sensitive data stored on its hybrid cloud environment that is accessed by authorized users across the globe. You need to implement a security solution that can use machine learning to protect your system from DDoS attacks. Which of these services should you use for this?

- A) Cloud Endpoints

✖
- B) Cloud Spanner

✖
- C) Cloud Armor

✔
- D) Cloud Trace

✖

Explanation

You should use Cloud Armor for this scenario. Cloud Armor is a service that protects websites and software applications from threats like web attacks and distributed denial of service (DDoS) attacks. It offers adaptive protection by detecting and mitigating large numbers of Layer 7 DDoS incidents using a locally trained machine learning (ML) system. Cloud Armor has support for both cloud and hybrid cloud environments and contains web application firewall (WAF) rules that are preset for protection against common web application threats and Open Web Application Security Project (OWASP) Top 10 cyberthreats. It also has integration with reCAPTCHA Enterprise for providing protection against bot attacks as well as rate limiting features to protect your application from illegal requests that may be flooding your instances.

You would not use Cloud Trace in this scenario. Cloud Trace is a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights into its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

You would not use Cloud Endpoints in this scenario. Cloud Endpoints provide a system for managing application programming interfaces (APIs) that enables monitoring, analyzing, and securing APIs as well as specifying quotas for them. You can monitor the key operational metrics for the APIs using Google Cloud Console and gain usage insights with BigQuery, Cloud Trace, and Cloud Logging. You can also utilize Cloud Endpoints Portal to create a website that API users can access to find documentation for the APIs and interact with them.

You would not use Cloud Spanner in this scenario. Cloud Spanner is a relational database service that is fully managed and ideal for critical business applications. You would use Cloud Spanner for provisioning a highly scalable database for online transaction processing (OLTP). This database will have high levels of availability and consistency worldwide.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe a cloud security approach and Google Cloud security benefits

References:
[Cloud.google.com > Google Cloud Armor](#)

Related Material

Question #13 of 60

Question ID: 1495284

Your company has adopted the DevOps philosophy of software development. As part of implementing DevOps, which of these will the company need to monitor?

- A) The time taken for cloud adoption

✖
- B) The returns made on investments

✖
- C) Employee satisfaction

✖
- D) System health and reliability

✔

Explanation

The system's health and reliability need to be monitored and measured as part of the DevOps philosophy. DevOps is an initiative that is geared toward increasing the speed of delivering software, making services more reliable, and creating ownership that is shared among all stakeholders. DevOps principles are used to enforce security best practices right from the beginning of each development project. Google's DevOps Research and Assessment (DORA) team has found certain key capabilities that an organization can leverage for improved performance and faster software delivery. These capabilities are related to the company's culture, measurement systems, technology, and processes.

Employee satisfaction is not measured as part of the DevOps philosophy. Employee satisfaction depends on various factors, which are not related to the processes and practices that comprise DevOps and its underlying frameworks. DevOps is focused on bringing technology, processes, and people together in a way that facilitates faster software development with greater stability and security.

The returns made on investments are not measured as part of the DevOps philosophy. The measurement of risk and return on investment (ROI) are aspects unrelated to creating an efficient framework for software development that yields stable and secure software delivered swiftly. Monitoring costs and measuring ROI are key functions of business teams and management, and are important for ensuring that the organization's financial performance is in line with the strategy and cost management principles of the company.

The time taken for cloud adoption is not measured as part of the DevOps philosophy. By adopting Google's Cloud Adoption Framework (CAF), a company can make a smooth transition to the cloud based on its current transformational phase. This is not related to creating a DevOps culture and utilizing best practices based on the DevOps system. The CAF is a way for an organization to assess its cloud readiness and find actions to prepare itself for successful cloud adoption. The CAF provides a structure of people, processes, and technologies that are involved in a company that is planning a cloud adoption.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Explain how IT operations need to adapt to thrive in the cloud

References:
[Cloud.google.com > DevOps](#)

Related Material

Question #14 of 60

Question ID: 1497137

You want to build greater safety and accountability into your company's AI products. To help you achieve this, you are looking for relevant tools that are part of Google's Responsible AI framework. Which tools would you use? (Choose two.)

- A) Looker

✖
- B) Cloud Trace

✖
- C) Model cards

✔
- D) Cloud Storage

✖
- E) TensorFlow open-source toolkit

✔

Explanation

Model cards, TensorFlow open-source toolkit, and Explainable artificial intelligence (AI) are tools used as a part of Google’s Responsible AI framework. Responsible AI tools allow you to inspect and analyze existing AI models to ensure safer products, increased customer trust, and ethically sound innovation. Explainable AI is a collection of frameworks and tools for understanding and interpreting machine learning (ML) model predictions and is integrated with various Google services. Model cards provide a structured way to organize ML Models’ key facts, limitations, and capabilities. TensorFlow is an open-source platform used for ML. TensorFlow can be utilized for integrating the processes required for Responsible AI into an existing workflow for ML.

Cloud Trace is not used as a part of the Responsible AI system, but it is used for tracing purposes. Cloud Trace gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

Cloud Storage is not used as a part of the Responsible AI system. It allows you to store and access data on Google Cloud using a Representational State Transfer (RESTful) service. It also provides features for security and data sharing. Cloud Storage can be used for Object Lifecycle Management (OLM), where data can be configured so that data is automatically moved to lower-cost storage classes when the data meets specified criteria like a period having passed or a new version of the data created. Cloud Storage also provides data redundancy through various locations for storage buckets. This provides fast response times and disaster recovery (DR) options.

Looker is not used as a part of the Responsible AI system. Looker is a platform for business intelligence that allows you to access intelligent data for making better business decisions. It also can be embedded into other applications for providing embedded analytics. Looker can be used to create alerts for proactively sending timely insights related to various business data. It also has dashboards that are interactive which allow for analyzing data in real-time through a trusted data model.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Responsible AI](#)

Related Material

Question #15 of 60

Question ID: 1495230

Your organization uses virtual machines for their main database processing application. Each time new capabilities need to be added to the application, the system needs to be brought offline, which causes significant downtime. What would you suggest as a more robust solution for this situation?

- A) Install updated hypervisors for running virtual machines

✖
- B) Use containers to compartmentalize the application

✔
- C) Upgrade the CPUs for the underlying infrastructure

✖
- D) Use high performance solid-state drives for storing application data

✖

Explanation

You should suggest using containers to compartmentalize the application in this scenario. By compartmentalizing an application using containers, you can then change parts of the application in isolation from the other parts. Containers are software packages that contain application code as well as dependencies, such as runtime environments and libraries, that are required for running an application. This allows containers to virtualize an OS and allows applications to be made portable across multiple systems. Google search, YouTube, and Gmail are all containerized applications. Containers allow you to:

- Separate responsibilities – This allows development teams to work on application logic and features and operations teams to focus on application deployment.
- Make workloads portable – This allows containers to be run from any OS like Windows, Mac, or Linux and on any system including virtual machines (VMs), physical servers, a developer’s laptop, on-premises systems, or a public cloud.
- Isolate applications – Containers perform OS-level virtualization of resources like memory, CPU, storage media, and networking systems which isolates the OS from any other applications.

A key difference between containers and VMs is that containers perform virtualization at the level of the OS while VMs perform virtualization at the level of hardware.

You would not use high performance solid-state drives (SSDs) for storing application data. The storage medium for the system is not a contributing factor in performing application compartmentalization, which this scenario requires. You can use a variety of storage media for your system which can include hard disk drives (HDDs) or SSDs depending on your specific requirements.

You would not install updated hypervisors for running VMs. Hypervisors allow VMs to optimally execute several applications on VMs concurrently. However, they do not have provisions for compartmentalizing applications.

You would not upgrade the CPUs for the underlying infrastructure. Changing the CPUs can increase the processing capabilities of the underlying system but this will not alter the application architecture which needs to be compartmentalized using container technology.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > What are Containers?](#)

Related Material

Question #16 of 60

Question ID: 1495941

Your company has various business services in use that require the procurement of third-party software. You need to do a security analysis of this approach. What can be the potential problems that can occur when using third-party software in an existing cloud deployment?

- A) Third-party software cannot handle important business services

✖
- B) Third-party software lacks the support systems available for cloud-based software

✖
- C) Third-party software reduces a company's innovation goals

✖
- D) Third-party software can be prone to cyberattacks

✔

Explanation

Third-party software can be prone to cyberattacks. Third-party software can have security issues in it, which can compromise an organization's data security and make the organization vulnerable to cyberattacks. There needs to be a comprehensive security architecture in place within a company's IT processes to check third-party software and monitor it on a regular basis to ensure there are no security issues.

Third-party software has ample support systems available through various channels that include vendor support, online forums, and support groups. Third-party software can be developed by companies or groups that have a strong grasp of cybersecurity challenges and take the necessary measures and perform development best practices to ensure the software is secure and error free. The security level of software depends greatly on the development methodology employed to create it as well as the level of security awareness of the team that created it. Some open-source frameworks have forums and collaborative groups where open knowledge sharing contributes to mitigating any existing issues.

Third-party software does not reduce a company's innovation goals. The level of innovation that a company can incorporate into its operation depends more on the scale to which it has embraced digital transformation and not on its reliance on third-party software. To reach a state of being fully innovative, an organization will first need to ensure that it has open knowledge sharing, decentralized teams, more autonomous decision making, and use of artificial intelligence (AI) systems by team members of varying technical skill.

Third-party software can handle important business services and serve various critical functions of an organization. The amount of capability available in third-party software varies across industries and vendors. Third-party software can be as powerful as cloud offerings and sometimes even exceed the current capabilities of a cloud software offering. You need to select the right third-party software for the task at hand and make the decision based on various factors, which include the security of the third-party software.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe a cloud security approach and Google Cloud security benefits

References:
[Cloud.google.com > Google security overview](#)

Related Material

Question #17 of 60

Question ID: 1495197

A company that develops database applications is looking for a Google Cloud service that will enable its applications to get better performance by incorporating the availability and scalability of Redis services without the associated complexities of managing the Redis deployments. Which service should you use in this scenario?

- A) Firestore**
- B) Cloud Bigtable**
- C) Memorystore**
- D) Datastore**
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Explanation

You should use Memorystore in this scenario. Memorystore for Redis is a scalable, fully managed service for Redis on Google Cloud that provides 99.9% availability using up to five read replicas that are distributed across multiple availability zones (AZs). Memorystore provides automation for complex Redis and Memcached tasks, which include ensuring high availability, creating failovers, and performing patching. It is compatible with the protocols of both Memcached and Redis. Memorystore also has provision for private IP addresses and virtual private cloud networks (VPCs) for data security and has integration with Identity and Access Management (IAM) for access control. Memcached provides in-memory caching for temporary data storage. Redis is a key-value NoSQL database using an in-memory data system for high performance. Redis supports sharding and uses persistent data, while Memcached does not.

Firestore is not the appropriate choice for this scenario. Firestore is a serverless document database that is scalable and fully managed. It enables fast development for applications for the Internet, mobile, and the Internet of Things (IoT) that can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features. It also allows you to customize various rules for security and data validation to protect your data. Firestore can be integrated with BigQuery and Firebase. Firestore is an upgrade to the older service called Datastore, which provides a NoSQL database that is scalable.

Dataprep is not the appropriate choice for this scenario. Dataprep is a data service for exploring, cleaning, and preparing data visually for the purpose of machine learning (ML) and analysis. It is an intelligent data service that is serverless, so the user does not need to deploy or manage the infrastructure. Dataprep suggests the data transformations that are ideal each time a User Interface (UI) input is detected, so the user does not need to write code for these applications. Dataprep also detects any anomalies that may exist in the data, including duplicates and missing value data. It helps the user avoid having to assess the quality of the data and proceed directly to exploratory data analysis.

Cloud Bigtable is not the appropriate choice for this scenario. Cloud Bigtable is a NoSQL database service that is fully managed and enables you to work with massive workloads for analytics and operations with availability at 99.999%. It is ideal for uses that include applications for IoT, finance, banking, personalized services, and advertising technology. Cloud Bigtable has a storage engine that allows for improved predictions in applications utilizing ML. It also has integration with tools for big data including Hadoop, Dataproc, and Dataflow.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Memorystore](#)

Related Material

Question #18 of 60

Question ID: 1495162

You work for a company that is considering revising its plan of procuring physical servers and infrastructure to using the Google Cloud platform instead.

Which of these should you suggest as a key benefit of using Google's Cloud computing platform?

- A) A pricing system that ensures both used and unused resources are paid for**
- B) Access of compute services exclusively through an on-premises networking system**
- C) Quick deployment of new applications without provisioning underlying infrastructure**
- D) Investment by the company in the latest computing technologies**
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Explanation

You should suggest quick deployment of new applications without provisioning underlying infrastructure as a key benefit of using Google's Cloud computing platform. This is because the infrastructure required for developing and deploying applications is managed by Google Cloud, which frees up a company to use its staff for creating applications needed to drive business growth.

Access of compute services exclusively through an on-premises networking system is not a key benefit of using Google's Cloud computing platform. This is because when using a cloud computing system, users can access computing resources using an Internet connection from anywhere.

Investment by the company in the latest computing technologies is not a key benefit of using Google's Cloud computing platform. The company does not need to make investments in new and emerging technologies because it is the cloud providers' expertise to provide the necessary services to the company using the latest technologies. This saves the company a lot of capital by not having to invest in trending technologies that may be out of use in a few years.

A pricing system that ensures both used and unused resources are paid for is not a key benefit of using Google's Cloud computing platform. This is because in cloud computing, a company only needs to pay for the resources it actually consumes and not for any unused resources. This is also referred to as a Pay-As-You-Go (PAYG) system.

The key benefits of cloud computing you need to know about include:

- Flexibility – Cloud computing is architected in a way that allows users to access cloud systems from any location using an Internet connection and have scalability for services as per their requirements.
- Efficiency – Cloud computing allows companies to deploy new applications quickly without needing to take on the overhead of the necessary infrastructure.
- Strategic value – Cloud service providers have access to the latest technologies, which customers can avail of as services. This removes the risk of clients having to invest in new technologies which may become obsolete in time.
- Security – Security risks on a cloud platform are relatively low compared to an equivalent data center for an enterprise. This is due to the strength of cloud security systems and teams.
- Cost effectiveness – Cloud computing ensures that companies only pay for resources consumed. This ensures that data centers do not need to be provisioned with more capacity to handle spikes in usage or organizational growth. This allows a company to have its staff work on strategic tasks which help grow the business.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why cloud technology is revolutionizing business

References:
[Cloud.google.com > What is Cloud Computing? > What are the benefits of cloud computing?](#)

Related Material

Question #19 of 60

Question ID: 1495229

You need to find the right cloud infrastructure option for migrating from a legacy system with specific licensing needs and running your workloads. Which Google service should you use when you need to have granular control over infrastructural resources, such as CPU and RAM, on a particular OS?

- A) Compute Engine
- B) Cloud Run
- C) App Engine
- D) Cloud Functions

Explanation

You should use Compute Engine for this scenario. You would use Compute Engine when migrating from a legacy system that has specific OS and licensing needs. Compute Engine provides the use of virtual machines (VMs) where you can specify the exact memory, CPU, disk storage, and graphics processing units (GPUs) you need, along with the exact OS and other applications you need to use.

You would not opt for App Engine in this scenario. App Engine provides a serverless platform that is fully managed for building and hosting full web applications. You can write a web application using a preferred language and then use App Engine to manage the tasks related to application and database scaling as well as performing updates to the application. App Engine is the preferred option for when you need to focus on coding a web application or a backend for a mobile app and leave the tasks of deployment and scaling to Google Cloud.

You would not opt for Cloud Functions in this scenario. Cloud Functions is best suited for when you need to run functional code based on an event or a trigger. Cloud Functions runs your code when events occur from various sources, such as Pub/Sub, Cloud Storage, and Hypertext Transfer Protocol (HTTP). For example: running a process for video transcoding automatically whenever a video is uploaded to Cloud Storage.

You would not opt for Cloud Run in this scenario. You can use Cloud Run for situations where you need to perform deployment of containerized applications that are written in your preferred coding language. Cloud Run provides support for HTTP, Hypertext Transfer Protocol Secure (HTTPS), and websockets. You can use Cloud Run for creating application programming interfaces (APIs), applications for processing data, webhooks, and websites. With Cloud Run, code can be packaged into a container and simply deployed without needing infrastructural scaling. Another related container technology is Google Kubernetes Engine (GKE), which allows you to create and manage Kubernetes clusters. Kubernetes provides an open-source system to automatically manage, deploy, and scale container-based applications. GKE provides detailed control over containers and nodes, enabling you to specify CPU, RAM, storage, GPUs, and network configurations, along with an Autopilot option for situations where there is not enough operational and developer support available.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Where should I run my stuff? Choosing a Google Cloud compute option](#)

Related Material

Question #20 of 60

Question ID: 1495195

A company has an on-premises PostgreSQL server database system and is looking for a cloud-based alternative that is fully managed so the company can remove overhead costs associated with replication, maintenance, and updates. Which of these solutions should you recommend to meet this requirement?

- A) Dataproc
- B) Cloud Data Fusion
- C) Cloud SQL
- D) Cloud Trace

Explanation

You should recommend Cloud SQL for this scenario. Cloud SQL provides a relational database system that is fully managed and works with SQL Server, PostgreSQL, and MySQL. It provides features like flags for configurations, rich extension collections, and ecosystems for developers. Cloud SQL brings down the maintenance costs of databases and provides reliable operations supported by a site reliability engineering (SRE) team that is available around the clock. It has automation built-in for tasks like managing storage capacity and provisioning for databases. It can also be integrated with Google Kubernetes Engine (GKE) and BigQuery while also providing observability of your data using Cloud SQL Insights.

You would not recommend Cloud Trace in this scenario because it is used as a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

You would not recommend Cloud Data Fusion in this scenario because it is a fully managed service that provides scalable integration for cloud-native data. It allows users to deploy Extract, Transform, and Load/Extract, Load, and Transform (ETL/ELT) data without requiring coding by using a visual interface. Cloud Data Fusion is created using the Centralized Data Analysis Platform (CDAP) data analytics platform which is open source.

You would not recommend Dataproc in this scenario because it is better suited for applications related to data lakes. It provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, perform ETL operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:

[Cloud.google.com > Cloud SQL](#)

[Cloud.google.com > What is Cloud SQL?](#)

Related Material

Question #21 of 60

Question ID: 1497143

You have an existing ML workflow, and you need to investigate how Vertex AI can be used in it. Which of these stages in your ML workflow can you manage using Vertex AI? (Choose two.)

- A) Data lake modernization

✗
- B) ML model training and evaluation

✓
- C) Asynchronous messaging between applications

✗
- D) System bottleneck identification

✗
- E) Dataset creation and data upload

✓

Explanation

You can manage Dataset creation, data upload, and ML model training and evaluation using Vertex Artificial Intelligence (AI). Vertex AI provides a singular application programming interface (API) platform to build and scale ML models swiftly with features for custom tooling and pre-training. It allows you to manage the AI and ML development lifecycle for a company. These stages of the ML workflow can be managed using Vertex AI:

- Dataset creation and data upload
- ML model training and evaluation
- Hyperparameter tuning
- Uploading and storing of ML models inside Vertex AI
- Deployment of the trained model onto an endpoint for prediction serving
- Sending of prediction requests to the endpoint and setting traffic splits
- Performing management of models and endpoints

You would not use Vertex AI for data lake modernization. You can perform data lake modernization, Extract, Transform, and Load (ETL) operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not use Vertex AI for system bottleneck identification. You can use Cloud Trace for such applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance.

You would not use Vertex AI for asynchronous messaging between applications. You would use Pub/Sub for such applications. Pub/Sub is a middleware for messaging that enables applications to send messages to each other asynchronously. An application designated as a “publisher” application sends messages to a topic, and all the applications that have subscribed to that topic will receive those messages.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:

[Cloud.google.com > Introduction to Vertex AI](#)

Related Material

Question #22 of 60

Question ID: 1497124

Which of these are characterized as key advantages of moving to a data-driven culture? (Choose two.)

- A) Easy access to analytics data for the business users of a company

✓
- B) Remove the need for data governance

✗
- C) Less reliance on fully managed services for data management

✗
- D) Creation of duplicate data for use within an organization

✗
- E) Use of a Zero Trust security model for data sharing

✓

Explanation

Easy access to analytics data for the business users of a company and the use of a Zero Trust security model for data sharing are both key advantages of moving to a data-driven culture. A data-driven culture gives data the topmost priority and forms the basis of its decision making for its strategies and teams. A data-driven culture incorporates people, processes, and technology, and ensures that a business:

- operates with trust
- ensures insights are democratized
- increases its agility
- applies business intelligence

Less reliance on fully managed services for data management is not an advantage of moving to a data-driven culture. This is because data can be used efficiently when using more managed integrated systems. This bridges any skill gaps that may exist for users and enables them to operate, scale, and manage data with high levels of agility. This allows for greater user collaboration and removes bottlenecks and silos in an organization.

The creation of duplicate data for use within an organization is not an advantage of moving to a data-driven culture. A data-drive culture seeks to create a single source of reliable data within an organization that can be accessed by users when they need it. This can be achieved by ingestion of data into a data warehouse and building data models with business intelligence systems, such as Looker. This makes relevant data available for all teams without issues related to redundancy and provisioning overhead.

Removing the need for data governance is not an advantage of moving to a data-driven culture. Data governance is an integral part of a data-driven culture and ensures that only the correct users can access relevant data and also ensures that data is safe throughout its lifecycle.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Describe the role of data in digital transformation and the importance of a data-driven culture

References:
[Cloud.google.com > Google's guide to building a data-driven culture](#)

Related Material

Question #23 of 60

Question ID: 1497129

Based on the Google AI Maturity scale, which of these indicates transformational maturity in AI automation for a company? (Choose two.)

- A) Automatically detecting data anomalies using regularly run data validation tasks**
- B)** Constantly updating ML models with fresh data using ML pipelines for continuous training
- C) Creating a system that combines ML systems development and operations**
- D)** Manually performing model training, validation, and data analysis
- E)** Performing ad hoc data analytics and ML tasks

Explanation

Transformational maturity in artificial intelligence (AI) automation is characterized by creating a system that combines machine learning (ML) systems development and operations, and automatically detecting data anomalies using regularly run data validation tasks. When continuous integration/continuous delivery (CI/CD) is applied to ML, it involves validation of ML models, data schemas, and code as part of CI, and it involves the deployment of ML models and continuous testing (CT) pipelines as part of the CD. To create a production system for ML, you would use ML Metadata, Container Registry Cloud Build, and Model Registry services.

Constantly updating ML models with fresh data using ML pipelines for continuous training does not indicate transformational maturity in AI automation for a company. This is a characteristic of strategic maturity. Strategic maturity is characterized by the standardization of processes for governing data flow within the data lifecycle, such as data ingestion to data analysis. This leads to more agility and smaller cycle durations. Because many of these ML models created at the strategic phase fail in the real world, there is a continuous training ML pipeline created to update models with fresh data. Strategic maturity features automation of pipelines to process and analyze data.

Performing ad hoc data analytics, ML tasks, and manually performing model training, validation, and data analysis do not indicate transformational maturity in AI automation for a company. These are characteristics of tactical maturity, which involves using ready-for-use AI tools, data visualization based on sheets, and tools for data wrangling that manually create systems for AI applications. All the steps, including data analysis, data preparation, and training of ML models and their validation are performed manually. It also features data scientists writing and running experimental code using Jupyter Notebooks to find an ML model that works.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Services.google.com > Google Cloud's AI Adoption Framework](#)

Related Material

Question #24 of 60

Question ID: 1495227

You want to create a cloud-based testing environment that duplicates various different hardware setups so that the testing team can perform various application tests within this isolated environment. Which of these technologies should you use for this requirement?

- A)** Containers
- B)** Physical computers
- C)** Serverless data warehouses
- D) Virtual machines**

Explanation

This is a use case for virtual machines (VMs). A VM is a virtual computer that runs operating systems and applications. Multiple VMs can run on a single physical computer and access various resources from a hypervisor. A hypervisor is software that creates an abstraction of all the computing resources of the physical computer and provides these resources to VMs running on the computer as required. The VM architecture provides the underlying framework in use for modern cloud computing. Compute Engine is Google's VM service, which offers VMs that can be customized. Some uses of VMs in enterprises include:

- Server consolidation – VMs can act as servers to host other VMs and so reduce sprawl by consolidating multiple computing systems into a single physical machine.
- Migration of workloads – VMs are flexible and portable, which enable swift workload migration for companies.
- Test environment creation – VMs can provide an isolated system that can be used for testing without affecting the surrounding systems.
- Disaster recovery – You can duplicate an existing system as a VM on the cloud, thus adding an additional layer of security to the system.
- DevOps – VMs can be activated, deactivated, and migrated smoothly, which facilitates flexible app development.
- Hybrid systems – VMs help you flexibly integrate existing on-premises legacy systems with a cloud environment to create a hybrid computing environment.

You would not use containers for this scenario. Containers are better suited for creating applications that need to be compartmentalized and made portable across multiple platforms. By compartmentalizing an application using containers, you can then change parts of the application in isolation from the other parts. Containers are software packages that contain application code as well as dependencies, such as runtime environments and libraries, that are required for running an application. This allows containers to virtualize an OS and allow applications to be made portable across multiple systems.

You would not use serverless data warehouses for this scenario. A serverless system like BigQuery would be suitable for a requirement where there needs to be very high levels of scalability for working with very large quantities of data. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence (BI), and geospatial analytics ready for use. It also has a serverless architecture that allows you to use SQL queries to gain key organizational data. BigQuery has a scalable analysis engine that has a distributed structure and allows you to perform queries on petabytes of data within minutes.

You would not use physical computers for this scenario. This is because procuring physical computers for creating testing environments is not as economical, feasible, or flexible as using a VM based system. A single physical computer has one hardware configuration with a specific CPU, RAM, and storage setup. Using several different VMs, you can have several hardware setups, which makes the process of testing an application across multiple platforms much simpler.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > What is a Virtual Machine?](#)

Related Material

Question #25 of 60

Question ID: 1495255

Your company offers data analytics services using resources on its on-premises systems as well as its cloud deployment. It is now looking for a way to host its APIs both on-premises and on Google Cloud. Which Google Cloud service should you recommend for this?

- A) Apigee hybrid**
- B) Dataproc**
- C) Cloud Trace**
- D) Cloud Backup and DR**
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Explanation
You should recommend Apigee hybrid for this scenario. With Apigee hybrid, you can place gateways near the source of application programming interface (API) traffic to use existing governance and security infrastructure while also benefiting from cloud-native services for monitoring and analytics. This is made possible by managing and controlling the runtime environment using Apigee hybrid. Apigee is used to build, secure, scale, and analyze APIs at any location with support for the complete API lifecycle. It provides strong systems of API control and visibility. With Apigee, you can increase the value of your digital assets by monetizing your API-based products.

You would not use Cloud Backup and Disaster Recovery (DR) in this scenario. Cloud Backup and DR is a service that provides a way to backup workloads from both Google Cloud and on-premises systems for data protection that is consistent as well as centralized. It allows you to have a single system to protect file systems, database instances, and virtual machines (VMs) for both VMware and Compute Engine. Using application-consistent backups that are incremental-forever, Cloud Backup and DR ensures low values for both the recovery point objective (RPO) and recovery time objective (RTO) of a company, ensuring swift recovery of business operations using backups.

You would not use Dataproc for this scenario. Dataproc provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, Extract, Transform, and Load (ETL) operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not use Cloud Trace in this scenario. Cloud Trace is a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Cloud.google.com > Apigee API Management](#)

Related Material

Question #26 of 60

Question ID: 1495294

Your company has a large Google Cloud deployment running multiple applications. It needs to find a way to ensure optimal system health and performance for its applications. Which Google Service should you use to gather relevant metadata, events, and metrics from various components of your applications?

- A) Cloud Vision**
- B) Cloud Spanner**
- C) Cloud Filestore**
- D) Cloud Monitoring**
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-
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Explanation

You should use Cloud Monitoring. Cloud Monitoring can be used to monitor the complete infrastructure of a cloud deployment. It provides a view into the uptime, health, and performance of cloud systems. For this, Cloud Monitoring gathers relevant metadata, events, and metrics from sources that include uptime probes, services, app instrumentation, alerts, notifications, and other components of applications. It allows you to stop issues from happening, and it allows you to identify trends. It also reduces overhead related to monitoring through a service that is integrated and provides metrics, dashboards, and alerts all in one place. Cloud Monitoring provides service level objective (SLO) monitoring and metrics collection for virtual machines (VMs) (using Ops Agent) and Kubernetes (using Managed Service for Prometheus).

You would not use Cloud Vision for this scenario. Cloud Vision is used by developers to understand image content through machine learning (ML) models accessed through an application programming interface (API). It provides image classification into various categories, object and face detection inside images, and identification of printed words inside of images.

You would not use Cloud Spanner for this scenario. Cloud Spanner is a relational database service that is fully managed and ideal for critical business applications. You would use Cloud Spanner for the provision of a highly scalable database for online transaction processing (OLTP). This database will have high levels of availability and consistency worldwide.

You would not use Cloud Filestore for this scenario. You would use this when you need a fully managed and highly available file service that has the capability of storing persistent data for shared workloads. You would use Cloud Filestore for enterprise-grade data applications that need storage with durability and persistence, which can be accessed using a Network File System (NFS).

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Identify Google Cloud solutions for cloud resource monitoring and application performance management

References:
[Cloud.google.com > Cloud Monitoring](#)

Related Material

Question #27 of 60

Question ID: 1495238

You are creating a plan for digitally transforming workloads for your company. What would be the purpose of using cloud-native applications during application modernization?

- A)** Using APIs for ensuring all business teams work at the same pace
- B) Partitioning workloads for failure protection**
- C)** Consolidating a single unified backend for redundancy
- D)** Avoiding cost overruns caused by application decoupling

Explanation

Using cloud-native applications allows for partitioning workloads for failure protection. The process of functional decomposition involves identifying individual components and their related systems inside of existing workloads. Then, you can create a decoupled architecture where workloads are broken into smaller parts and scaled independently. This can help you provide higher availability of services as well as protection from failures by maintaining several read replicas of a service within a system that is distributed.

Consolidating a single unified backend for redundancy is not a purpose of application modernization. You would instead spread backends and other services over several availability zones (AZs) and use read replicas and backups to ensure that the system is not affected by failures. You may need to maintain certain services in a specific geographical location for compliance, and you may need to distribute other services globally for low latency content distribution to users.

Avoiding cost overruns caused by application decoupling is not a purpose of application modernization. Performing decoupling of applications as part of application modernization ensures that services that scale up or down more frequently are separated from other services that are more stable. This way, you cut back on costs associated with scaling up of services that are not required or are unrelated.

Using application programming interfaces (APIs) to ensure all business teams work at the same pace is not a purpose of application modernization. You would use APIs to perform application decoupling for existing business services. This is a form of functional decomposition that breaks down an existing legacy or “monolith” application into independent applications. With APIs for various business services, multiple service users and teams can use these services at their own pace. This system also provides greater visibility into the usage of these services in the company.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand modernizing applications with Google Cloud

References:
[Inthecloud.withgoogle.com > CIO Guide to Application Modernization](#)

Related Material

Question #28 of 60

Question ID: 1495228

You need to guide your company toward using more cloud-native services to create greater business value. Which of these are characteristics of serverless computing using the cloud? (Choose two.)

- A)** Operational costs are higher
- B) Suitable only for predictable scaling needs**
- C) Infrastructure is provisioned by the cloud service provider**
- D)** Code is run by actual servers
- E)** Computing resources need to be managed by users

Explanation

Two characteristics of serverless computing include:

- Infrastructure is provisioned and scaled by the cloud service provider (CSP)
- Code is run by actual servers

Serverless computing is a cloud computing model that allows you to create and execute code without needing to manage the underlying servers or pay for any infrastructure that is unused.

Some serverless Google technologies include:

- Application development – You can use Cloud Functions and App Engine for serverless hosting of applications. You can use Cloud Datastore and Firebase for NoSQL serverless databases.
- Analytics – You can use Cloud Storage and Pub/Sub for serverless data analytics work and Dataflow for processing global scale datasets.
- Microservices – Google serverless systems help develop microservices based systems by bringing down the cost of infrastructure by dynamic scaling. Using fully managed serverless systems also reduces operational costs, which removes the need for companies to perform server management. Also, using serverless systems like Pub/Sub can help avoid synchronous call chains by creating precomputed results asynchronously for microservices.

Computing resources do not need to be managed by users. This is done by the CSP and allows developers to focus on creating business value instead of performing resource provisioning. Also, with the automatic and dynamic nature of resource provisioning using serverless systems, applications developed on serverless systems can perform massive levels of scaling.

Serverless systems do not create higher operational costs. This is because when provisioning servers for database and computing applications is done by the CSP, the operational costs to provision and manage massive server clusters reduce substantially.

Serverless systems are suitable not only for predictable scaling needs but excellent for unpredictable workloads. Serverless systems like Google's Cloud Functions and App Engine both offer dynamic scaling based on request volumes without the need for performing server management by the developer. You can use Cloud Functions for executing small chunks of code triggered by Hypertext Transfer Protocol (HTTP) requests or cloud events. Since the billing is only done per request, this can result in significant savings for traffic patterns that are uneven or unpredictable. Similarly, you can use App Engine for larger sections of code like for a Website.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Serverless Architecture](#)

Related Material

Question #29 of 60

Question ID: 1497154

Your company's cloud deployment is used by several teams and departments. You need to find an efficient way to ensure that the operating costs of the cloud deployment stays within the set budgetary limit specified by management. Which of these actions should you take to accomplish this?

- A) Manage inventory for on-premises systems

B) Use cloud resource labels and hierarchies

C) Use organizational policies for resource access

D) Set resource quotas and perform throttling
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Explanation
You should set resource quotas and perform throttling. Setting resource quotas and performing throttling needs to be done to ensure that cost overruns do not occur due to resource utilization that exceeds the specified budget. Using Google Cloud tools, you can perform automated throttling and capping of resource consumption as well as broadcast budget notification alerts on company communication channels, such as Slack. The most important capabilities for cost management on the cloud include:

- Maintaining costs within budgetary limits
- Finding ways to save on IT costs
- Performing forecasting for IT expenditure
- Maintaining consistency in IT costs and keep deviations limited

You would not use organizational policies for resource access in this scenario. You would use these policies to gain more control over your cloud costs. The purpose of using organizational policies is to specify permissions for users to perform spending on Google Cloud based on their role in the organizational hierarchy. This way, you control the ability of specific users to view costs and perform administrative functions using Cloud Identity and Access Management (Cloud IAM) for cloud resources.

You would not use cloud resource labels and hierarchies in this scenario. You can achieve greater accountability in your cloud deployment for financial governance by using cloud resource labels and hierarchies. Using resource hierarchies simplifies the process of allocating costs to specific departments to ensure accountability. This also provides insights into the return on investment (ROI) of the investment you have made on the cloud.

Managing inventory for on-premises systems is not a way of implementing budgetary control on a cloud deployment. An on-premises system and its inventory are independent of cloud resources. The financial management of on-premises systems is not related to the financial governance systems that need to be implemented to manage costs on a cloud deployment.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > A guide to financial governance in the cloud](#)

Related Material

Question #30 of 60

Question ID: 1495221

A game development company has been developing games from a legacy on-premises data center and is now considering making a digital transformation. How would the company benefit from modernizing its infrastructure with cloud technology?

- A) Massive scalability by using serverless technology

B) Ensuring all resources are paid for regardless of usage

C) Making significant investments in new emergent technologies

D) Delegating application security to the cloud service provider
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Explanation

Using cloud-based technologies like serverless computing, a company can achieve massive levels of scalability, which may not be feasible using traditional legacy systems. For example, BigQuery has a scalable analysis engine that has a distributed structure and allows you to perform queries on petabytes of data within minutes.

Delegating application security to the cloud service provider (CSP) is not a benefit of using cloud technology. Security of applications and all the data stored in the cloud is the responsibility of the customer and not the CSP. The CSP is responsible for the security of its data centers and of the cloud.

Paying for all resources, regardless of usage, is not a benefit of using cloud technology. Cloud computing ensures that companies only pay for resources consumed. This ensures that data centers do not need to be provisioned with more capacity to handle spikes in usage or organizational growth, and allows a company to have its staff work on more strategic tasks that help grow the business.

Making significant investments in new emergent technologies is not a benefit of using cloud technology. CSPs have access to the latest technologies, which customers can avail of as services. This removes the risk of the company having to make large investments into new technologies, which may become obsolete in time.

Some distinct benefits of cloud technology are outlined below:

- Security – Through a shared responsibility model and best practices, you can ensure greater safety of your operations using Google Cloud.
- Reliability – This helps you ensure your workloads can perform correctly, handle failures, and automatically fix any issues that may arise. This also covers data backups, fault isolation, and disaster scenario preparation.
- High Availability – This ensures that your applications are always ready to serve customers swiftly with almost no downtime. A measure of availability is five 9s, which corresponds to a system being available 99.999% of the time.
- Elasticity – This is the ability to scale your computing resources to match consumer demand without having to perform resource provisioning manually or guess future capacity needs.
- Agility – This is the ability to swiftly deploy new applications by removing unnecessary operational overhead so that the development team can focus on creating and innovating products.
- Pay-as-you-go (PAYG) pricing – This is a flexible payment option where a cloud consumer only pays for resources consumed.
- Scalability – This is using the right technology to easily make your application meet growing demands on a massive scale using serverless technologies.
- Global Reach – Using edge locations and data centers around the world, Google Cloud’s global infrastructure can improve user experience with low latency for end users, regardless of their location on the globe.
- Economy of scale – This is having lower PAYG pricing due to accumulated cloud usage from a very large number of cloud customers.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > What is Cloud Computing? > What are the benefits of cloud computing?](#)

Related Material

Question #31 of 60

Question ID: 1495244

You work for a company that runs a photo sharing website using Google Cloud for its operations. You want image resizing to be performed automatically whenever images are saved on Cloud Storage. Which of these Google Cloud services will allow you to do this?

- A) Cloud Run

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- B) App Engine

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- C) GKE

✖
- D) Cloud Functions

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Explanation

You should use Cloud Functions for this scenario. Cloud Functions is best suited for when you need to run functional code based on an event or a trigger. Cloud Functions runs your code when events occur from various sources like Pub/Sub, Cloud Storage, and Hypertext Transfer Protocol (HTTP). An example is running a process for video transcoding automatically whenever a video is uploaded to Cloud Storage.

You would not use Cloud Run for this scenario. You can use Cloud Run for situations where you need to perform deployment of containerized applications which are written in your preferred coding language. Cloud Run provides support for Hypertext Transfer Protocol (HTTP), Hypertext Transfer Protocol Secure (HTTPS), and websockets. You can use Cloud Run to create application programming interfaces (APIs), applications for processing data, webhooks, and websites. With Cloud Run, code can be packaged into a container and simply deployed without needing infrastructural scaling.

You would not use App Engine for this scenario. App Engine allows you to create and host your applications on the same platforms Google uses for several of its own applications. With App Engine, you can build and deploy applications quickly and manage them easily. It provides scalability without the issues related to backups, patching, and hardware provisioning. App Engine provides support for web application development using programming languages like Python, Ruby, C#, Node.js, Go, and PHP.

You would not use Google Kubernetes Engine (GKE) for this scenario. GKE allows you to create and manage Kubernetes clusters. Kubernetes provides an open-source system to automatically manage, deploy, and scale container-based applications. GKE provides detailed control over containers and nodes, enabling you to specify CPU, RAM, storage, GPUs, and network configurations, along with an Autopilot option for situations where there is not enough operational and developer support available.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand modernizing applications with Google Cloud

References:
[Cloud.google.com > Cloud Functions](#)

Related Material

Question #32 of 60

Question ID: 1495193

A company that is in the digital publishing domain has approached you to help them find a suitable technology for building a data warehouse that they can use for performing analysis and reporting with scalability that is seamless. Which Google Cloud service should you recommend for this requirement?

- A) Cloud Storage

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- B) Cloud Trace

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- C) DataFlow

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- D) BigQuery

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Explanation

You should recommend BigQuery for this scenario. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence (BI), and geospatial analytics ready for use and has a serverless architecture allowing you to use SQL queries for gaining key organizational data. BigQuery has a scalable analysis engine that has a distributed structure. This allows you to perform queries on petabytes of data within minutes.

You would not recommend DataFlow for this scenario because it is not a data warehousing service but a service for performing data processing. DataFlow is a fully managed service that automates the process of provisioning and managing processing resources. It provides the highest possible resource utilization by performing horizontal autoscaling for various worker resources.

You would not recommend Cloud Storage for this scenario because it is an object storage service and not a data warehousing service. Cloud Storage can be used for Object Lifecycle Management (OLM), where data can be configured so that data is automatically moved to lower-cost storage classes when the data meets specified criteria like a period having passed or a new version of the data created. Cloud Storage also provides data redundancy through various locations for storage buckets. This provides fast response times and disaster recovery (DR) options.

You would not recommend Cloud Trace in this scenario as it is used as a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > What is BigQuery?](#)

Related Material

Question #33 of 60

Question ID: 1495186

An e-commerce company is expanding its workforce and is working on performing data analysis to find ways to grow its business. You need to use a suitable service to improve decision making and productivity in the organization by providing key insights to a growing number of users. Which tool should you use for this requirement?

- A) Dataproc

B) Cloud Trace

C) Dataflow

D) Looker
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Explanation

You should use Looker. Looker is a platform for business intelligence (BI) that allows you to access intelligent data for making better business decisions. It can be embedded into other applications for providing embedded analytics. Looker can be used to create alerts for proactively sending timely insights related to various business data. It also has dashboards that are interactive, which allow for analyzing data in real-time through a trusted data model.

You would not use Dataproc for this scenario. Dataproc provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, Extract, Transform, and Load (ETL) operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not use Dataflow for this scenario. Dataflow is a fully managed service that automates the process of provisioning and managing processing resources. It provides the highest possible resource utilization by performing horizontal autoscaling for various worker resources.

You would not use Cloud Trace in this scenario as it is not for BI, but a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Looker](#)

Related Material

Question #34 of 60

Question ID: 1495946

You work for a data analytics company that uses the Google Cloud infrastructure for performing all its data warehousing and business intelligence workloads. You need to perform an evaluation of the cloud infrastructure and investigate uptimes and response rates. Which Google Cloud tool should you use for this?

- A) Cloud Monitoring

B) Cloud Vision

C) Cloud Trace

D) Cloud Debugger
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- Explanation**

You should use Cloud Monitoring for this scenario. Cloud Monitoring can be used to monitor the complete infrastructure of a cloud deployment. It provides a view into the uptime, health, and performance of cloud systems. For this, Cloud Monitoring gathers relevant metadata, events, and metrics from sources that include uptime probes, services, app instrumentation, alerts, notifications, and other components of applications. It allows you to stop issues from happening, and it allows you to identify trends. It also reduces overhead related to monitoring through a service that is integrated and provides metrics, dashboards, and alerts all in one place. Cloud Monitoring provides service level objective (SLO) monitoring and metrics collection for virtual machines (VMs) (using Ops Agent) and Kubernetes (using Managed Service for Prometheus).

You would not use Cloud Trace for this scenario. Cloud Trace is a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights into its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

You would not use Cloud Vision for this scenario. Cloud Vision is used by developers to understand image content through machine learning (ML) models accessed through an application programming interface (API). It provides image classification into various categories, object and face detection inside images, and identification of printed words inside of images.

You would not use Cloud Debugger for this scenario. Cloud Debugger is better suited for performing an analysis of an application's state at a specific location of code without the need to halt or slow down application requests. Cloud Debugger does this by creating a connection between the production data of your application to the source code.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Identify Google Cloud solutions for cloud resource monitoring and application performance management

References:
[Cloud.google.com > Cloud Monitoring](#)

Related Material

Question #35 of 60

Question ID: 1497149

A bank has an on-premises data center that it needs to maintain to ensure regulatory compliance. It also has several applications that it uses for internal purposes as well as for serving its customers. What should you use to provide a single portal for visibility of their nodes and clusters in their cloud and on-premises workloads that also manages large volumes of Kubernetes clusters inside their applications?

- A) App Engine

B) Firestore

C) Anthos

D) Cloud Functions
- ✖

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Explanation

You should use Anthos for this scenario. Using Anthos, you can create and manage an application and then run it on both on-premises systems and on the cloud. With Anthos, you can perform migrations straight from virtual machines (VMs), and create and deploy apps using Google Kubernetes Engine (GKE), VMs, and landing zones that are serverless. Anthos allows you to use its platform for deploying both on-premises and cloud-native applications to create a global fleet that has consistency in its operations. Using containers and serverless services, Anthos lets you grow productivity. By maintaining security systems at every stage of the application's lifecycle, Anthos provides protection for both a software supply chain and application.

You would not use App Engine for this scenario. App Engine allows you to create and host your applications on the same platforms Google uses for several of its own applications. With App Engine, you can build and deploy applications quickly and manage them easily. It provides scalability without the issues related to backups, patching, and hardware provisioning. App Engine provides support for web application development using programming languages like Python, Ruby, C#, Node.js, Go, and PHP.

You would not use Cloud Functions for this scenario. Cloud Functions is best suited for when you need to run functional code based on an event or a trigger. Cloud Functions runs your code when events occur from various sources, such as Pub/Sub, Cloud Storage, and Hypertext Transfer Protocol (HTTP). For example: running a process for video transcoding automatically whenever a video is uploaded to Cloud Storage.

You would not use Firestore for this scenario. Firestore is a serverless document database that is scalable and fully managed. It enables fast development for applications for the internet, mobile, and the Internet of Things (IoT), which can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features. It also allows you to customize various rules for security and data validation to protect your data. Firestore can be integrated with BigQuery and Firebase. Firestore is an upgrade to the older service called Datastore, which provides a NoSQL database that is scalable.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand modernizing applications with Google Cloud

References:
[Cloud.google.com > Anthos](#)

Related Material

Question #36 of 60

Question ID: 1495898

Your company is considering using a hybrid cloud setup where you will use several public clouds and your existing on-premises servers.

Which cloud migration method would you use if you need to refactor an existing application and modernize it before moving it to a cloud deployment?

- A) Rip and replace

B) Lift, tinker, and shift

C) Improve and move

D) Lift and shift
- ✖

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Explanation

You would use the “Improve and move” method in this scenario. This method involves taking an existing application and first refactoring and modernizing it. Then the application is moved to a public cloud system. This movement to a public cloud is an ideal situation for companies that want to create new products and services using new technologies.

You would not use the “Lift and shift” method in this scenario. This method involves the migration of an application that was previously in a private cloud environment to a public cloud without making any key modifications. This approach is used by companies that want minimal disruption to their operations, and want to migrate to the cloud while keeping their operating costs as low as possible.

You would not use the “Rip and replace” method in this scenario. This method involves replacing an existing application completely with a new cloud-based version. This approach is used for situations where modernizing the existing application is simply not a feasible alternative due to cost or technology related limitations.

You would not use the “Lift, tinker, and shift” method in this scenario. This method is for situations where you need to modify parts of an application to gain the benefits of a cloud deployment without changing the underlying core of the application.

When considering migration strategies for a hybrid cloud deployment, you also need to be aware of the utilities of a hybrid cloud system:

- Well-paced modernization – A hybrid cloud approach allows you to migrate your applications to a cloud at a speed that is comfortable for your organization. This way you can change the IT infrastructure of your organization over a period of time.
- Regulatory compliance – Each industry has certain stipulations that specify where an application is allowed to operate. Using a hybrid cloud deployment ensures that you keep the use of your application and data within regulatory compliance laws.
- On-premises applications – Some of your applications may need to be run on mainframe computers on-premises. Moving these systems to a cloud deployment may not be feasible. For such regulated applications, a hybrid cloud deployment may be suitable.
- Applications at edge locations – Some industries and applications require the use of edge computing to provide low latency. This could be for systems like retail kiosks or telecommunication networks. Hybrid cloud systems allow you to run such specific applications at remotely placed edge locations.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why cloud technology is revolutionizing business

References:
[Cloud.google.com > What is a Hybrid Cloud? > How-to hybrid](#)

Related Material

Question #37 of 60

Question ID: 1497161

A company has a sketching app used by millions of users around the world, and now it needs to make some changes to the application’s front end. Which would be the recommended course of action as per Site Reliability Engineering (SRE) best practices?

- A)** Do not make the release until automation for toil is removed
- B)** Do not make the release until there is a zero chance of errors
- C) Release the modification to a small subset of users for testing**
- D)** Release the modification to all users to ensure coverage

Explanation

As per Site Reliability Engineering (SRE) best practices, it is recommended that the company release the modification to a small subset of users for testing. A key SRE best practice is ensuring that changes are released gradually with testing performed for the changes. Some SRE best practices for production servers include:

- Control failures – Ensure failures don't impact the functionality and allow systems to work using a known configuration and only use new data or configurations after they have been sanitized and validated by a human user.
- Make rollouts progressive – All rollouts and updates that need to be performed should be performed in stages and monitored throughout. The changes should be applied to a small section of traffic across time frames and geographies. This allows for problems to be detected and mitigated.
- Specify service level objectives (SLOs) from a user’s perspective – This involves setting measures for performance and availability using metrics that are applicable to a user. An example of this is the latency and error rates of a client application.
- Maintain a balance between error budgets and innovation – The system’s reliability and speed of performing innovation need to be in balance with error budgets, which are the acceptable failure levels of a specified service. An example is a 0.01% error budget for a service with 99.99% availability. This implies that a team can plan to add new features and innovation to a service only if the service has not already availed its error budget for the month, which includes errors and downtime.
- Performing monitoring – System monitoring for health can yield outputs that include pages, tickets, and logging, and require human intervention depending on the severity of a reported issue.
- Do blameless postmortems – This ensures that systems are improved in their design so that complete groups of possible errors that could arise are mitigated. It is done by not blaming people and focusing on making changes to technology and processes.
- Plan for capacity – This includes making plans to handle planned and unplanned outages while ensuring it does not affect a user beyond an acceptable limit. It involves performing load testing to find resource-to-capacity relationships and validating demand forecasts with actual demands until the two match consistently.
- Checking for overloads – This involves making sure that services still produce acceptable results even if overloaded so that users can continue using them. It may require certain features of a service to be halted in an overload situation, while the main service features are still made available.
- Limit operational tasks for SRE teams – This is to make sure that SRE teams do not dedicate more than half of their time to performing operational tasks (toil) that should be sent to development teams. For this, you can ensure there are regular meetings between SRE and development teams. Toil comprises work that may be manual, nonenduring, automatable, repetitive, and tactical.

You must not release the modification to all users to ensure coverage. As per SRE best practices, until a rollout or modification is tested and verified by a small set of users across various geographies and time frames, it must not be made available to the full set of users. This is to ensure any bugs or errors are mitigated before the change reaches all users,

Not making the release until there is a zero chance of errors is not a part of SRE best practices. Errors and failures are a natural part of the SRE process, but failure needs to be dealt with in a controlled manner. As per SRE best practices, you need to ensure that a steady state of the service is maintained so that any failure that occurs is contained and does not cause a catastrophic failure of the entire service.

Not making the release until automation for toil is removed is not an SRE best practice. On the contrary, a key goal of SRE is to ensure toil automation takes place. Toil automation removes the need for performing manual, repetitive work, which does not have an enduring value.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Explain how IT operations need to adapt to thrive in the cloud

References:
[SRE: A Collection of Best Practices for Production Services](#)

Related Material

Question #38 of 60

Question ID: 1495908

You work as a cloud consultant for a manufacturing firm that wants to use cloud-based business intelligence (BI) tools to identify issues in its production lines. Which of these would you need to do to ensure that your company makes the best use of its BI tools?

- A) Normalize and clean the data for BI**
- B)** Opt for an on-premises BI tool for future scalability
- C)** Encourage all users to use open source BI tools

D) Keep data for BI siloed in different systems

Explanation

You would need to normalize and clean the data for business intelligence (BI). For data to be useful for BI, you need to ensure it is high quality, updated, and cleaned. BI is a process of using technology to collect and study organizational data for making strategic organizational decisions. This involves collecting the data into a storage repository like a data warehouse and using tools to perform data analysis on the stored data. Looker is a BI tool offered by Google Cloud. Some examples of BI applications include analysis of the shopping preferences of customers, costs of company operations, and sales data. BI is also useful for comparing a company's operations versus pre-established standards and baselines. There are three categories of BI tools:

- On-premises – These are used inside an on-premises data center with data warehouses also on-premises. They lack the scalability available in cloud-based BI tools.
- Open source – These are affordable tools and can help save on the costs of infrastructure when used as cloud-based variants. However, they need considerable expertise and coding skills to be used.
- Cloud-based – These are highly effective tools for working with data in high volumes and streaming formats. They are also economical as the technical expertise and infrastructural overhead required for their use is the responsibility of the cloud service provider (CSP).

You would not keep data for BI siloed in different systems as this does not allow users access to all of the data required for BI purposes. Siloed data is distributed across multiple systems and cannot be accessed by other systems due to outdated software or stringent user access permissions. This makes key data needed for BI become inaccessible. Using an Extract, Transform, and Load (ETL) tool can help consolidate the data from across multiple systems needed for BI applications.

You would not encourage all users to use open source BI tools. You need to use managed tools for BI as much as possible because it reduces the complexity of using the tools and ensures that a lack of programming expertise or technical know-how does not become a performance bottleneck in the BI process. Open source BI tools require technical knowledge and may not be suitable for all users inside an organization. You would instead opt for a managed cloud-based BI tool, such as Looker.

You would not opt for an on-premises BI tool for future scalability because on-premises systems are not as scalable as cloud-based solutions. For accommodating future growth and working with massive datasets, you need a cloud-based solution that allows you access to managed data warehouses and BI tools with various features built in for ease of use.

You need to know of some key ways an organization can utilize BI tools, including:

- Reporting of key data for creating strategy and initiatives for the business
- Mining data through massive datasets for identifying trends and patterns
- Providing visualization of data for easy comprehension of complex data
- Analyzing real-time and streaming data, which may be gathered from sources, such as Internet of Things (IoT) sensors and stock market data
- Using predictive modeling and analysis for forecasting future trends using historical data
- Measuring business performance by analyzing sales data and customer feedback

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > What is Business Intelligence?](#)

Related Material

Question #39 of 60

Question ID: 1497160

Your company uses Google's GKE for automatically managing and deploying container-based applications. As per the shared responsibility model, which of these is the responsibility of the customer?

- A) The operating system of nodes
- B) Workloads containing docker files
- C) Encryption of data at rest
- D) The control pane

Explanation

The customer is responsible for workloads that contain docker files, application code, container images, Identity and Access Management (IAM) and Role-Based Access Control (RBAC) policies, and pods. The customer is also responsible for the nodes that are running workloads. This also includes the software that may be installed on nodes as well as any changes to the configuration of the nodes. Node updating is also the responsibility of the customer.

The operating system of nodes, encryption of data at rest, and the control pane are the responsibility of Google. This also includes the cloud infrastructure, which comprises OS, networking, storage, firmware, and hardware. Google is also responsible for the encryption of data at rest and in transit. The operating system in use on nodes, which can be Ubuntu or Container-Optimized OS (COS). is the responsibility of Google. Google Kubernetes Engine (GKE) can automatically deploy upgrades to the images. This is separate from the OS that is running inside of containers. Also, GKE manages the control pane that comprises master virtual machines (VMs), IAM authentication systems, cluster certificate authority (CA), and the application programming interface (API) server. The necessary patching and upgrades needed for these components are supported by a service level objective (SLO). The control pane is the part of the Kubernetes system that controls how communication with a cluster is performed and how user states are implemented. Google hardens this control pane to protect against vulnerabilities.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe a cloud security approach and Google Cloud security benefits

References:
[Cloud.google.com > Exploring container security: the shared responsibility model in GKE](#)

Related Material

Question #40 of 60

Question ID: 1495166

Your company is looking at ways to change its operation and resource utilization so that it excels at delivering the maximum possible value to customers. For this, it is looking into performing digital transformation using cloud technology.

Which of these is a key part of the journey to digital transformation?

- A) Segregating customer feedback from product development activities
- B) Guiding business decisions using insight from data analytics tools
- C) Keeping team collaboration focused within each company department
- D) Using legacy on-premises systems for quickly deploying new global applications

Explanation

Guiding business decisions using insight from data analytics tools is a key part of the journey to digital transformation. Experimentation with new technologies and concepts can help a company gain more insight into market needs and customer preferences. This experimentation can then be analyzed using advanced data analytics tools to make the right business decisions for growth.

Segregating customer feedback from product development activities is not a key part of the journey to digital transformation. This is because customer feedback needs to be incorporated into product development efforts and not segregated from them so the company can create offerings that better serve customers. Focusing on customers and their preferences through analysis of customer usage data can help create products that provide more value to customers and lead to greater business growth.

Keeping team collaboration focused within each company department is not a key part of the journey to digital transformation. Digital transformation involves using tools to share knowledge and collaborate across a company's organizational boundaries. Team collaboration and knowledge sharing eventually leads to more autonomy in decision making and creates a more dynamic workforce.

Using legacy on-premises systems for quickly deploying new global applications is not a key part of the journey to digital transformation. Digital transformation involves using modern technology that has the flexibility to enable you to deploy applications onto the cloud swiftly and manage them efficiently.

Digital transformation involves utilizing new technology to change the design and definition of a company's relationships with its partners, customers, and employees. Digital transformation involves using new digital systems including private, public, and hybrid clouds to change existing culture, work processes, and customer experiences with an aim to meet the changing dynamics in businesses and markets. Some key benefits of digital transformation include:

- Modernized infrastructure – This includes using serverless systems, containers, and global cloud networks for an efficient and effective IT infrastructure.
- Better data management – This include using new systems and tools to manage massive amounts of data which originate from various devices and sources.
- Stronger insights – This covers using advanced business analytics tools to provide useful data insights.
- Faster collaboration – This involves dismantling the traditional team silo concept where a team is cut off from other teams and instead using new tools that allow teams to collaborate across various global locations swiftly. Faster collaboration helps create value and produce results for customers more quickly.
- Finding business solutions – This involves using new systems, processes, and technologies for solving business problems and identifying ways to improve how businesses interact and work with their customers.
- Savings on costs – This involves using tools for making processes more efficient and identifying the high and low performance aspects of a business accurately.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why cloud technology is revolutionizing business

References:
[Cloud.google.com > What is Digital Transformation > What are the benefits of digital transformation?](#)

Related Material

Question #41 of 60

Question ID: 1495239

Your company is a software development firm that has been using a legacy on-premises system and now wants to leverage cloud technology to reduce bugs caused by differences in the development and production environments. What kind of change pattern for application modernization should you opt for in this use case?

- A) Keep specific personnel dedicated to testing tasks**
- B) Perform containerization of all services gradually**
- C) Combine multiple services into a single monolith**
- D) Use a manual release process for all systems**

Explanation

You should perform containerization of all services gradually. When a development environment is too different from an existing production environment, there can be various bugs and errors generated. A solution to this is to containerize all services gradually over time. This creates a greater level of control and predictability over the environments for development, production, and staging. You start by containerizing a single service. After you have gained confidence using certain processes and tools for containerization, you can proceed to containerize all services. This is a key aspect of application modernization.

You would not use a manual release process for all systems. A release process for software that is performed manually is prone to errors and is slower than an automated system. A key DevOps practice is continuous integration and continuous delivery (CI/CD), which specifies that a controlled and systematic approach needs to be used when making changes to ensure system stability and avoiding configuration drift. A public cloud system offers features like CI/CD for rapid and automated development and updates for applications. You would first use a simple CI model and then employ CD. After some time, you would apply the CI/CD model to all existing services. Then you can add additional stages to your CI/CD pipeline gradually.

You would not keep specific personnel dedicated to testing tasks. You would instead opt for automated testing systems. Manual processes for testing are prone to human errors. For this reason, when performing application modernization, an automated approach is recommended. By utilizing a basic CI/CD pipeline for your containers, you can have an automated system for testing and deploying your software.

You would not combine multiple services into a single monolith. Decoupling is a key part of modernizing applications that allows for greater flexibility in making changes to parts of the application independently. This is a form of functional decomposition that breaks down an existing legacy or “monolith” application into independent applications. One way to establish decoupling is using services like Pub/Sub. Pub/Sub is a middleware for messaging that enables applications to send messages to each other asynchronously. An application designated as a “publisher” application sends messages to a topic, and all the applications that have subscribed to that topic will receive the messages. This way, senders are decoupled from receivers in an architecture that is referred to as a “loosely-coupled” architecture, which forms the basis of a microservices-based architecture.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand modernizing applications with Google Cloud

References:
[Inthecloud.withgoogle.com > CIO Guide to Application Modernization](#)

Related Material

Question #42 of 60

Question ID: 1497148

Your company wants to minimize retrieval costs for any data that may be archived prematurely. You are looking for an automated cloud-based solution for moving data to the appropriate storage tier based on last access times. Which Google Cloud feature or service should you use to help you do this?

- A) Cloud Backup and DR**
- B) Cloud Build**
- C) Autoclass**
- D) Storage Insights**

Explanation

You should use Autoclass. This is a feature of Cloud Storage that shifts stored objects to more warm or cold storage classes based on predefined policies. This shifting is performed based on the last access time for the objects. This removes the need to perform data shifting manually and ensures cost optimization and predictability of prices through an automated system that is policy driven.

Storage Insights is not the correct option in this scenario. You would use Storage Insights to gather insights on the objects stored in Cloud Storage, which you can use to perform actions. With Storage Insights, you can gain information about trillions of objects stored on Cloud Storage and know the buckets the objects are stored in. You can also integrate this with BigQuery and create dashboards to visualize the insights into this stored data.

Cloud Backup and Disaster Recovery (DR) is not the correct option in this scenario. This service provides a way to backup workloads from both Google Cloud and on-premises systems for data protection that is consistent as well as centralized. It allows you to have a single system to protect file systems, database instances, and virtual machines (VMs) for both VMware and Compute Engine. The incremental-forever system used by Cloud Backup and DR ensures that duplicate data does not need to be stored and results in reduced cost of storage and faster recovery times.

Cloud Build is not the correct option in this scenario. Cloud Build is used to build, test, and deploy systems using a continuous integration and continuous delivery (CI/CD) platform, which is serverless. It allows a company to integrate its existing legacy systems and tools into a modern build system. This is made possible using third-party applications and customized steps for builds. Cloud Build also has vulnerability scanning features for CI/CD that prevents deployment of any images that are unsafe.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > New storage innovations to drive your next-gen applications](#)

Related Material

Question #43 of 60

Question ID: 1495901

You need to suggest a scaling strategy for a company that wants to move from its maturity level of strategic cloud adoption to one that is transformational. Which of these actions will you suggest for this requirement?

- A) Perform each modification when issues are found

✖
- B) Perform each modification using a manual approach

✖
- C) Perform each modification using a programmatic approach

✔
- D) Keep the system immutable with no modifications

✖

Explanation

You will suggest that the company perform each modification using a programmatic approach. This is because taking a transformational approach involves the use of automation. Automation ensures that changes are made in a constant and consistent manner with low risks to the system. This also ensures that changes can be rolled back, if necessary, to maintain a safe system configuration.

You would not perform each modification using a manual approach. This is not a transformational approach, but a tactical approach that involves changes being made infrequently and can pose high risks for the company. This approach is not recommended.

You would not perform each modification when issues are found. This is not a transformational approach that ensures that risks and problems are mitigated before they actually occur.

You would not keep the system immutable with no modifications. This is because a transformational approach requires rethinking and redefining existing systems and relationships for leveraging new technologies to facilitate operational excellence and business growth.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why it is critical for businesses to adopt new technology

References:
[Services.google.com > The Google Cloud Adoption Framework](#)

Related Material

Question #44 of 60

Question ID: 1495275

Your company is using Google Cloud and needs to find a way to incorporate its existing employee accounts in Active Directory with Google Cloud. What will you need to do to accomplish this?

- A) Federate user identities between Cloud Identity and Active Directory

✔
- B) Configure a lifecycle policy for data objects on the cloud

✖
- C) Make a new Active Directory account for each employee

✖
- D) Maintain Google Cloud identities for each employee manually

✖

Explanation

You will need to federate user identities between Cloud Identity and Active Directory. Google Cloud Identity is used for managing access and authentication. You can use Cloud Identity to manage Google identities in the company using automation. Cloud Identity allows you to link the lifecycles of Google identities to the identity management systems that are already functioning in an organization. The federation of user identities between Active Directory ensures that Google identities are maintained in an automated manner. By provisioning a user in Cloud Identity and Active Directory, you can enable single sign-on (SSO) for authenticating the user. SSO is a system in which a user can utilize one set of credentials and log into multiple independent systems.

You would not maintain Google Cloud identities for each employee manually. This is not an optimal way of managing identities as employees already have an account with an identity store like Active Directory. Manually maintaining Google identities would add large overhead for management. Instead, opting for identity federation is a more appropriate approach in this scenario.

You would not configure a lifecycle policy for data objects on the cloud. This does not help manage access for users on Google Cloud, which is what the scenario requires. Configuring a lifecycle for data objects is a way of setting up storage in a way that suits your company's data storage needs. Google Cloud Storage offers various storage classes, which include standard, nearline, coldline, and archival. By moving data to less expensive storage classes, such as the archival class, you can save on costs for storing data that is accessed rarely. By using a lifecycle policy, you can automatically ensure that data objects change their storage class based on conditions you specify, which can be the age of the object.

You would not make a new Active Directory account for each employee. This is because the existing employee Active Directory accounts can be successfully incorporated into Google Identities using identity federation. Google also provides the Google Cloud Directory Sync tool that synchronizes key groups and users regularly from Active Directory to Cloud Identity.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe a cloud security approach and Google Cloud security benefits

References:
[Services.google.com > Google Cloud security foundations guide > Authentication and authorization](#)

Related Material

Your company is looking into the security implications of moving its data to the Google Cloud Platform. As per the shared responsibility model, which of these are the responsibility of Google?
(Choose two.)

- A) Patching the guest operating system on a VM

✗
- B) Access rights to Cloud Storage objects

✗
- C) Security of servers running Compute Engine instances

✓
- D) Data center security

✓
- E) Encryption keys

✗

Explanation

Data center security and the security of servers running Compute Engine instances are both the responsibility of Google. The cloud service provider (CSP) takes on the responsibility of the cloud infrastructure, which includes the data center and its servers.

Access rights to Cloud Storage objects, encryption keys, and patching the guest operating system on a virtual machine (VM) are all related to data stored on the cloud, which means they are the customer's responsibility.

It is important to note that in the shared responsibility model, the customer is responsible for the security within the cloud, and the CSP is responsible for the security of the cloud. Google (the CSP) is responsible for the security of the cloud, including the data center, physical servers, and services. The security of the Google customer's workload is the responsibility of the customer. To summarize, the following are the responsibility of Google:

- Data center security
- Virtualization infrastructure
- Software and hardware infrastructure
- Networking infrastructure
- Global infrastructure, including Availability Zones, Regions, and Edge locations

The Google customer is responsible for securing all the data that they create and store in the cloud, including:

- Patching and configuring operating systems running on Compute Engine instances
- Setting up security groups
- Administering user accounts and permissions

However, the responsibilities can also depend on the Google service that is being used. For example, when using Cloud Spanner, Google is responsible for providing automated failover, redundancy, backups, and disaster recovery. Google also performs automated patching of the database engine. Similarly, the customer is responsible for the security of the code that is run using Cloud Functions. The customer is also responsible for patching software on Compute Engine instances.

When using Google Kubernetes Engine (GKE), the customer is responsible for workloads that contain docker files, application code, container images, Identity and Access Management (IAM) and Role-Based Access Control (RBAC) policies, and pods. The operating system in use on nodes, which can be Ubuntu or Container-Optimized OS (COS), is the responsibility of Google. GKE can automatically deploy upgrades to these images. This is separate from the OS that is running inside of containers.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe a cloud security approach and Google Cloud security benefits

References:
[Cloud.google.com > Shared responsibilities and shared fate on Google Cloud](#)

Related Material

Your company has a large on-premises data center that has taken a considerable amount of investment to set up. The company is now looking toward growing its business through the use of a hybrid cloud.

Which of these would be a consequence of using a hybrid cloud solution?

- A) Reduced ROI by the addition of a public cloud

✗
- B) Increased latency for end users

✗
- C) Reduced maintenance of on-premises servers

✗
- D) Increased privacy and compliance for applications

✓

Explanation

You can increase the privacy and compliance of your applications when using a hybrid cloud solution. This is because you can specify the position of your application in relation to the hybrid cloud computing model. This helps you regulate your application, ensuring they fall within compliance requirements and ensure the necessary privacy controls for them.

Reduced maintenance of on-premises servers is not a consequence of using a hybrid cloud solution. While a hybrid cloud solution can allow you to migrate on-premises servers to the cloud and have the cloud provider maintain the hardware for the servers, maintenance for existing on-premises servers will be unaffected by a hybrid cloud solution.

A reduced return on investment (ROI) by the addition of a public cloud does not occur as you increase your computing capabilities through the addition of a public cloud provider to your existing on-premises systems. This can be done without an increase in your existing data center costs leading to an improvement in ROI.

Using a hybrid cloud solution does not increase latency for end users. When you have applications that are distributed at various remote locations, you can greatly benefit from using a hybrid cloud solution. Hybrid computing occurs near the end user's location and leads to reduced levels of latency.

The three main cloud deployment models you should know about include:

- Cloud-based (also known as Cloud native) – In this deployment, applications are fully functional in a cloud and have no on-premises portions.
- On-premises (also known as private) – In this deployment, all the infrastructure and applications that are part of the cloud or virtualization system are running on systems that are inside a company's on-premises data center.
- Hybrid – In this deployment, infrastructure and applications are shared between an on-premises environment and a cloud environment. A hybrid cloud model is used by companies that want to take advantage of the scalability and security offered through a public cloud while maintaining data on-premises for regulatory compliance needs.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why cloud technology is revolutionizing business

References:
[Cloud.google.com > What is a Hybrid Cloud? > What are hybrid cloud benefits?](#)

Related Material

Your company is looking for a serverless platform that is fully managed for running containers individually. What is the most efficient way to write code for an application and deploy it as a container with no additional overhead required?

- A) Compute Engine

B) Cloud Functions

C) App Engine

D) Cloud Run
- ✖

✖

✖

✔

Explanation

You should use Cloud Run for this scenario. You can use Cloud Run for situations where you need to perform deployment of containerized applications that are written in your preferred coding language. Cloud Run provides support for Hypertext Transfer Protocol (HTTP), Hypertext Transfer Protocol Secure (HTTPS), and websockets. You can use Cloud Run for creating application programming interfaces (APIs), applications for processing data, webhooks, and websites. With Cloud Run, code can be packaged into a container and simply deployed without needing infrastructural scaling. The apps containerized with Cloud Run can be written in Ruby, Python, Go, Java, Node.js, or .NET.

You would not use Cloud Functions for this scenario. Cloud Functions is best suited for when you need to run functional code based on an event or a trigger. Cloud Functions runs your code when events occur from various sources like Pub/Sub, Cloud Storage, and Hypertext Transfer Protocol (HTTP). An example is running a process for video transcoding automatically whenever a video is uploaded to Cloud Storage.

You would not use Compute Engine for this scenario. Compute Engine provides virtual machine (VM) computing that has scalability and flexibility. It allows you to utilize Cloud Tensor Processing Units (TPUs), Graphics Processing Units (GPUs), and CPUs to perform analytics and processing on a large scale using Google's technological infrastructure.

You would not use App Engine for this scenario. App Engine allows you to create and host your applications on the same platforms Google uses for several of its own applications. With App Engine, you can build and deploy applications quickly and manage them easily. It provides scalability without the issues related to backups, patching, and hardware provisioning. App Engine provides support for web application development using programming languages like Python, Ruby, C#, Node.js, Go, and PHP.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand modernizing applications with Google Cloud

References:
[Cloud.google.com > Cloud Run](#)

Related Material

You need to create a system that can provide intuitive and high quality conversations for customers through voicebots and chatbots. What Google technology would you select for this need?

- A) Dialogflow

B) Media Translation API

C) Document AI

D) Visual Inspection API
- ✔

✖

✖

✖

Explanation

You would use Dialogflow in this scenario. Dialogflow provides a suite for the development of conversational apps with voice and text capabilities, which include voicebots as well as chatbots. Dialogflow provides cross-platform connectivity to existing apps for the Internet of Things (IoT), iOS, or Android, and for telephony platforms, including Cisco, Genesys, and Avaya. It can also be used for digital platforms, including Slack, Facebook Messenger, and Google. The Dialogflow Customer Experience Edition (CX) enables the advanced development of chatbots and voicebots. It contains a visual platform for building bots, tools for modularizing bots, support for advanced interactive voice response (IVR), as well as tools for versioning and collaboration. It has been optimized and scaled for use at an enterprise-level.

Document Artificial Intelligence (AI) is not the appropriate choice for this scenario. You would use Document AI for the classification and extraction of structured data from existing documents. This can enable you to find key insights and automate organizational practices. You can use Human-in-the-Loop AI for workflow systems and a user interface to verify the information extracted from documents using Document AI.

Media Translation application programming interface (API) is not the appropriate choice for this scenario. You would use Media Translation API for automatic translation of audio between languages. This can also be done in real-time, if required. You can use Media Translation API for programmatic translation of audio inside of your applications. Media Translation API is based on the open source Google Remote Procedure Call (gPRC) framework.

You would not use Visual Inspection API for this scenario. You would use Visual Inspection API for training and deploying AI models for automatic detection, classification, and localization of any anomalies that are identified in images. This helps increase the quality of production in various industries and enables development of advanced analytics.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Dialogflow](#)

Related Material

A shopping company has several self-service terminals at various locations around the country that its customers can use to make payments and get information on discounts. Using this deployed hardware, the company also wants to leverage a new virtual application that will service customer requests. Why will the company need to use APIs with this system?

- A) To allow customers to access the Internet from the terminals

B) To perform data lifecycle management using object storage

C) To change key hardware components in the terminals

D) To allow the new application to communicate with the check-out hardware
- ✖

✖

✖

✔

Explanation

The company will need to use application programming interfaces (APIs) to allow the new application to communicate with the check-out hardware. With Cloud APIs, you can automate workflows using whichever programming language you prefer and use Representational State Transfer (REST) calls and client libraries as required. APIs can facilitate communication between legacy systems like check-out hardware and modern virtual customer service applications.

Performing data lifecycle management using object storage is not related to this scenario. Cloud Storage can be used for Object Lifecycle Management (OLM), where data can be configured so that data is automatically moved to lower-cost storage classes when the data meets specified criteria, such as a period of time passing, or a new version of the data being created. Cloud Storage also provides data redundancy through various locations for storage buckets. This provides fast response times and disaster recovery (DR) options.

Changing key hardware components in the terminals cannot be done using APIs. Changes to hardware components can be done by technicians and the department that handles hardware control and maintenance. APIs are a way for applications to communicate with each other and share services.

Allowing customers to access the Internet from the terminals is not the reason why you need to use APIs with this system. Customers should not be allowed to use the terminals for Internet access as it can pose a significant security risk to the system. A self-service terminal needs to be protected and secured from Internet access for safety.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Cloud.google.com > Cloud APIs](#)

Related Material

Question #50 of 60

Question ID: 1497135

Your company is in the process of adopting Google AI technology for its business growth. You are evaluating fairness and security in Google's AI and machine learning differentiators. Which of these is a recommended best practice for Google's stipulated responsible AI?

- A) Conduct short-term fixes to all issues for swift problem solving**
- B) Keep all the features in your ML model, including redundant ones**
- C) Perform engagement with multiple users in the development process**
- D) Train models to create automatic inferences based on detected correlations**

Explanation

Performing engagement with multiple users in the development process is a recommended best practice for Google's stipulated responsible artificial intelligence (AI) framework. This is part of a best practice of utilizing a design approach that is centered on actual users. The effectiveness of an AI system's predictions and decisions can be gauged from the experiences of actual users. Incorporating inputs from a wide range of users and use cases helps enrich the project with a range of perspectives from users. This eventually benefits the users that actually utilize the technology. The system needs to have enough built-in disclosures to users so that a satisfying user experience can be achieved.

Keeping all the features in your machine learning (ML) model, including redundant ones, is not a recommended best practice for Google's stipulated responsible AI. The recommended practice is to use the ML model, which is simple and meets goals for performance without using redundant features. The raw data used by an ML model needs to be analyzed, and if the model uses sensitive data, then that data should be understood within privacy mandates for the data. The data needs to be checked for errors, bias, and training-serving skews to ensure the optimal performance of the model.

Training models to create automatic inferences based on detected correlations is not a recommended best practice for Google's stipulated responsible AI. You need to ensure that the ML model does not make casual inferences based on correlations that it detects. You also need to be aware of the limitations of the model based on its training data's scope and range. Communicating these limitations to users can help with getting useful feedback about the AI system from users.

Conducting short-term fixes to all issues for swift problem solving is not a recommended best practice for Google's stipulated responsible AI. Short-term fixes may save time but do not offer solutions that are beneficial in the long-term. For this reason, it is important to create a balance between using short-term fixes and long-term solutions. Also, you need to compare the candidate ML model with the deployed version to identify the ways they differ, and to understand how the update will alter the user experience and system performance of the AI application.

Some additional best practices include using metrics like user feedback, false negatives and positives, product health reports, and system performance data for assessing monitoring and training of the ML model. Also, testing the system with both integration tests as well as unit tests to ensure that the performance of the AI system is as expected. This involves studying how the individual components of the ML model work with other components of the application. Also, continuous monitoring of the model needs to be done to ensure the model incorporates feedback from the real-world and user experiences.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Ai.google/responsibilities > Responsible AI practices](#)

Related Material

Question #51 of 60

Question ID: 1495242

Your company uses a hybrid cloud architecture with several of its applications needing to run on its on-premises data center as well as on Google Cloud. You need to recommend a service that can meet this requirement. Which service should you recommend?

- A) Compute Engine**
- B) Anthos**
- C) App Engine**
- D) Firestore**

Explanation

You should recommend Anthos. Using Anthos, you can create and manage an application and then run it on both on-premises systems and on the cloud. With Anthos, you can perform migrations straight from virtual machines (VMs), create and deploy apps using Google Kubernetes Engine (GKE), VMs, as well as landing zones that are serverless. Anthos allows you to use its platform to deploy both on-premises and cloud-native applications to create a global fleet that has consistency in its operations. Using containers and serverless services, Anthos lets you grow productivity. By maintaining security systems at every stage of the application's lifecycle, Anthos provides protection for both a software supply chain and an application.

You would not recommend App Engine. This cannot be used on-premises because it is a cloud computing platform. App Engine allows you to create and host your applications on the same platforms Google uses for several of its own applications. With App Engine you can build and deploy applications quickly and manage them easily. It provides scalability without the issues related to backups, patching, and hardware provisioning.

You would not recommend Compute Engine. This cannot be used on-premise because it is Infrastructure as a Service (IaaS). Compute Engine provides VM computing that has scalability and flexibility. It allows you to utilize Cloud Tensor Processing Units (TPUs), Graphics Processing Units (GPUs), and CPUs to perform analytics and processing on a large scale using Google's technological infrastructure.

You would not recommend Firestore. Firestore is a serverless document database which is scalable and fully managed. It enables fast development for applications for the internet, mobile, and Internet of Things (IoT), which can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features. It also allows you to customize various rules for security and data validation to protect your data. Firestore can be integrated with BigQuery and Firebase. Firestore is an upgrade to the older service called Datastore, which provides a NoSQL database which is scalable.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand modernizing applications with Google Cloud

References:
[Cloud.google.com > Anthos](#)

Related Material

Question #52 of 60

Question ID: 1497126

You are looking for a RESTful service for storage and access of your data on Google Cloud, as well as a way to move the data across various costing tiers based on access frequency. Which of these services meets this requirement?

- A) Looker

✖
- B) Persistent Disk

✖
- C) Cloud Storage

✔
- D) Cloud Filestore

✖

Explanation

Cloud Storage is the correct service to use in this scenario. Cloud Storage allows you to store and access data on Google Cloud using a Representational State Transfer (RESTful) service. It also provides features for security and data sharing. Cloud Storage can be used for Object Lifecycle Management (OLM), where data can be configured so that data is automatically moved to lower-cost storage classes when the data meets the specified criteria, such as a set period of time or when a new version of the data is created. Cloud Storage also provides data redundancy through various locations for storage buckets. This provides fast response times and disaster recovery (DR) options.

Persistent Disk is not the correct choice for this scenario. You would use Persistent Disk when you need a block storage service that is durable and offers high performance. Persistent Disk has storage capability utilizing Solid State Drives (SSDs) and Hard Disk Drives (HDDs), which you can attach to an instance. This instance can run in Google Kubernetes Engine (GKE) or Compute Engine.

Cloud Filestore is not the correct choice for this scenario. You would use Cloud Filestore when you need a fully managed and highly available file service that has the capability of storing persistent data for shared workloads. You would use Cloud Filestore for Enterprise-grade data applications that need storage with durability and persistence, which can be accessed using a Network File System (NFS). You can also use Cloud Filestore for storage systems that need a file system that is Portable Operating System Interface (POSIX) compliant.

Looker is not the correct choice for this scenario. You would use Looker when you need a platform for business intelligence (BI) that allows you to access intelligent data for making better business decisions. It can also be embedded into other applications to provide embedded analytics. You can use Looker to create alerts and proactively send timely insights that relate to various business data. It also has dashboards that are interactive, which allow you to analyze data in real-time through a trusted data model.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Cloud Storage](#)

Related Material

Question #53 of 60

Question ID: 1495183

Your company is creating a new e-commerce portal and wants to begin the process of data classification and management for it. You need to identify all the possible structured and unstructured data you will have to work with. Which of these is an example of structured data?

- A) Descriptions of the products written by vendors

✖
- B) Customer ratings for each product

✔
- C) Pictures of products featured on the site

✖
- D) Reviews written for products by customers

✖

Explanation

Customer ratings for each product are an example of structured data. This is because a product rating can be a numerical value or score. Credit card numbers and product names are also examples of structured data. Structured data is data that is organized and quantitative. This allows for easy storage and management using a database. Databases provide an organized way of storing data as tables that can be accessed using a computing system, and can ingest large amounts of data from various sources. The consistency and accuracy levels of data stored in a database comprise the transaction integrity.

The other options are not examples of structured data. They are unstructured data. Product reviews and descriptions both comprise text files, and product pictures are image files. Image files and documents from word processing applications are examples of unstructured data. Unstructured data comprises data that is qualitative and is stored as objects. This allows the data to be stored in its original format along with key metadata that provides ways to uniquely identify the data. Metadata is data about stored objects.

Some additional terms you need to know include:

- Data point – This represents a chunk of information, which can be customer purchases.
- Dataset – These are an aggregation of several data points, which belong to a particular category like “transactions by customers”.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for data management

References:
[Services.google.com > Cloud Digital Leader Study Guide](#)

Related Material

Question #54 of 60

Question ID: 1497144

The developers in your company need to use a programmatic approach for working with and deploying machine learning models. Which of these tools can be used for managing jobs and endpoints with Vertex AI?

- A) REST API**
- B) Cloud Data Fusion**
- C) Cloud SQL**
- D) BigQuery**
-
-
-
-

Explanation

You can use Representational State Transfer (REST) application programming interfaces (APIs) for interacting with Vertex Artificial Intelligence (AI). The Vertex AI REST API includes various RESTful services that enable you to manage jobs, machine learning (ML) models, and endpoints. The API can also be used to create predictions through models on Google Cloud. There are various tools for interacting with Vertex AI, including:

- REST API – These are used for managing endpoints and jobs.
- Google Cloud console – This provides a user interface for managing models, jobs, datasets, and endpoints as well as access to monitoring and logging tools.
- Cloud Client libraries – These provide optimization of the developer environment by providing support for specific languages for making calls to Vertex AI API.
- Deep Learning Virtual Machine (VM) images and containers – These include VMs that are optimized for ML jobs with ML frameworks pre-built and Docker containers with pre-installed tools and frameworks for data science.

You would not use BigQuery for this scenario. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and can be utilized for analytics. It has features like ML, business intelligence, and geospatial analytics ready for use. It has a serverless architecture allowing you to use SQL queries to gain key organizational data. BigQuery has a scalable analysis engine that has a distributed structure. This allows you to perform queries on petabytes of data within minutes.

You would not use Cloud Data Fusion for this scenario. Cloud Data Fusion is a fully managed service that provides scalable integration for cloud-native data. It allows users to deploy Extract, Transform, and Load/Extract, Load, and Transform (ETL/ELT) data without requiring coding by using a visual interface. Cloud Data Fusion is created using the Centralized Data Analysis Platform (CDAP) data analytics platform, which is open source. This allows for the portability of data pipelines and integration for multi-cloud and hybrid cloud deployments, which enables users to remove existing solos and access previously inaccessible data.

You would not use Cloud SQL for this scenario. Cloud SQL provides a relational database system that is fully managed and works with SQL Server, PostgreSQL, and MySQL. It provides features like flags for configurations, rich extension collections, and ecosystems for developers. Cloud SQL brings down the maintenance costs of databases and provides reliable operations supported by a site reliability engineering (SRE) team that is available round the clock. It has automation built-in for tasks like managing storage capacity and provisioning for databases.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Introduction to Vertex AI](#)

Related Material

Question #55 of 60

Question ID: 1495285

A software development company is using Google Cloud for its development environment. It is now considering using DevOps. What is the purpose of incorporating DevOps into a production environment? (Choose two.)

- A) Increase software stability**
- B) Perform less frequent testing**
- C) Increase in-person processes**
- D) Make deployments quicker**
- E) Lessen the need for monitoring tools**
-
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-
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-

Explanation

Increasing software stability and making deployments quicker are both goals of using DevOps. DevOps is an initiative that is geared toward increasing the speed of delivering software, making services more reliable, and creating ownership that is shared among all stakeholders. DevOps principles are used to enforce security best practices from the beginning of each development project. Google's DevOps Research and Assessment (DORA) team has found certain key capabilities that an organization can leverage for improved performance and faster software delivery. These capabilities are related to the company's culture, measurement systems, technology, and processes.

Increasing in-person processes is not the purpose of DevOps. Using automation is a key feature of the DevOps philosophy, which seeks to create an automated model for application development that results in more error-free operations. The automated approach also helps ensure that changes can be rolled back, if required, and system stability is ensured using safe configurations.

Performing less frequent testing is not the purpose of DevOps. A key DevOps practice is continuous integration and continuous delivery (CI/CD), which specifies that a controlled and systematic approach needs to be used for making changes to ensure system stability and avoid configuration drift. A public cloud system offers features like CI/CD for rapid and automated development and updates for applications. You would first use a simple CI model and then employ CD. After some time, you would apply the CI/CD model to all existing services. Then you can add additional stages to your CI/CD pipeline gradually. Continuous testing (CT) is a part of CD, which ensures testing occurs regularly and in sync with delivery.

Lessening the need for monitoring tools is not the purpose of DevOps. The system health and reliability need to be monitored and measured as part of the DevOps philosophy. For this, various monitoring tools need to be used. Monitoring helps understand the performance of applications and map resource consumption to specific code areas of the application for further optimization.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Explain how IT operations need to adapt to thrive in the cloud

References:
[Cloud.google.com > DevOps](#)

Related Material

Question #56 of 60

Question ID: 1495188

An organization is building a modern, unified analytics platform using Google Cloud. It must decide between using a data lake or a data warehouse for its data analytics needs. Which of these would be an appropriate business use case for using a data warehouse?

- A)** There is a need to explore insights using existing resources
- B) Known datasets need to be analyzed and preset questions answered**
- C)** There is no clarity on the kinds of analysis that may need to be done on the data
- D)** There is a need to discover data across various data types

Explanation

When known datasets need to be analyzed and preset questions answered then a data warehouse would be the appropriate choice for an organization. A data warehouse is a massive repository that contains a company's data gathered from various different sources. This data includes current data as well as data that is historical. A data warehouse is used for business intelligence (BI) activities like complex analytics which can lead to making informed management decisions. The sources of data for a data warehouse include customer relationship management (CRM) systems, automation systems for marketing, applications for transactions, and log files from applications. A data warehouse can be used for performing analysis in an ad hoc manner or for creating custom reports.

The other options are all characteristics of business use cases that can benefit from using a data lake:

- There is a need to discover data across various data types
- There is a need to explore insights using existing resources
- There is no clarity on the kinds of analysis that may need to be done on the data

A data lake allows you to store both structured, semi-structured, and unstructured data from multiple sources in a single place. This data can come from a variety of sources, including multimedia, social media, hard disks, servers, Internet of Things (IoT) devices, clickstream, databases, and transactional information. A data lake provides security and scalability and enables companies to perform ingestion of data from various sources including edge computing, cloud, or on-premises systems. The data can be stored in its original format and processed as a batch or in real-time. This data can then be analyzed using tools like Python, SQL, and R.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Services.google.com > Build a modern, unified analytics data platform > Data warehouse or data lake?](#)

Related Material

Question #57 of 60

Question ID: 1497133

You need to implement a system for ensuring data quality in prediction accuracy for deployed ML models. How would you do this?

- A) Use continuous integration and delivery for ML models**
- B)** Maintain accurate ML models without retraining
- C)** Manually update each ML model with fresh data
- D)** Use known data for ML models without regular updating

Explanation

You would use continuous integration and delivery for machine learning (ML) models for ensuring data quality in prediction accuracy for the deployed ML models. ML needs to have some quality control measures implemented for the best results. These measures include processes to evaluate models after training and to validate them pre-deployment. Additionally, there needs to be logging and monitoring systems in place so that the systems health can be checked and notifications issued in the event of any anomaly. When continuous integration/continuous delivery (CI/CD) is applied to ML, it involves validation of ML models, data schemas, and code as part of CI, and deployment of ML models and continuous testing (CT) pipelines as part of CD. To create a production system for ML, you would use ML Metadata, Container Registry Cloud Build, and Model Registry services.

You would not manually update each ML model with fresh data. This is not a suitable approach because manual updating can lead to errors and is not a feasible solution in an enterprise that may have a massive number of deployed ML models that need regular updates of their data as well as implementation of new ML concepts. An automated pipeline-based approach would be ideal for updating ML models. This approach is used in transformational maturity, which involves orchestration and automation of ML and data pipelines end-to-end.

You would not use known data for ML models without regular updating. This is because ML models need to be updated due to changes in the data that characterizes their operating environment. Failure to update ML models with fresh data can lead to ML models failing in deployment. Using CT as part of CD ensures that the ML models are fully tested and validated pre-deployment and re-trained regularly.

You would not maintain accurate ML models without retraining because ML models need to be trained continuously to have the most accurate predictions. This is to ensure that the models are accurate and can respond to real world data correctly. Deploying a CT pipeline as part of a larger CD system ensures that the ML models are retrained regularly. This ensures that all ML models currently deployed into production are updated with fresh data, which can capture trending patterns as well as applications of the latest ML concepts.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Services.google.com > Google Cloud's AI Adoption Framework](#)

Related Material

Question #58 of 60

Question ID: 1495191

Your company has various multi-cloud and hybrid cloud environments that store its data. You need to find a way to populate this data lake with data from all of these sources. Which service should you use to perform data ingestion for the data lake?

- A) Cloud Data Fusion**

✔
- B) Dataproc**

✘
- C) Cloud Storage**

✘
- D) Cloud Trace**

✘

Explanation

You should use Cloud Data Fusion for the process of data ingestion into the data lake. Cloud Data Fusion is a fully managed service that provides scalable integration for cloud-native data. It allows users to deploy Extract, Transform, Load/Extract, Load, Transform (ETL/ELT) data without requiring coding by using a visual interface. Cloud Data Fusion is created using the Centralized Data Analysis Platform (CDAP) data analytics platform, which is open source. This allows for the portability of data pipelines and integration for multi-cloud and hybrid cloud deployments that enable users to remove existing solos and access previously inaccessible data. Additionally, Cloud Data Fusion provides several preset transformations that can be used for stream and batch processing tasks. This boosts collaboration and productivity by allowing connections and transformations to be shared among users.

You would not use Cloud Storage for data ingestion. This is because it is an object storage service and not a data ingestion service. Cloud Storage can be used for Object Lifecycle Management (OLM), where data can be configured so that data is automatically moved to lower-cost storage classes when the data meets specified criteria like a period having passed or a new version of the data created. Cloud Storage also provides data redundancy through various locations for storage buckets. This provides fast response times and disaster recovery (DR) options.

You would not use Dataproc for data ingestion. Dataproc provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, ETL operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not use Cloud Trace for data ingestion. Cloud Trace is used as a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Cloud Data Fusion](#)

Related Material

Question #59 of 60

Question ID: 1495902

Your company needs to assess its readiness for adopting the Google Cloud platform. For this, you need to assess its cloud readiness. Which theme of the Google Cloud Adoption Framework covers the ability of a company to leverage services that are native to the cloud to reduce overhead in its IT operations?

- A) Lead**

✘
- B) Scale**

✔
- C) Learn**

✘
- D) Secure**

✘

Explanation

The correct option is scale. This is the theme that covers the level to which a company uses cloud services, provisions workloads, performs automation, and maintains the applications it needs for its business. This theme covers how a business uses native cloud technologies, like serverless computing and automation, to improve its operations and scalability.

The other options are incorrect as they do not measure a company's depth of involvement in leveraging cloud-native systems to improve its operations. The other options are also a part of the themes of the Google Cloud Adoption Framework (CAF) and address different aspects of the cloud readiness of a company. The CAF is a way for an organization to assess its cloud readiness and find actions to prepare itself for successful cloud adoption. The CAF provides a structure of people, processes, and technology that are involved in a company planning a cloud adoption. There are three components to Google's CAF:

- Three maturity phases
- Four cloud adoption themes
- Seventeen epics

A component of the CAF is the four themes:

- Learn – This measures the learning systems in place for an organization to upskill its IT workforce. It also includes the ability of a company to leverage partners who are skilled in cloud technology to augment its existing workforce.
- Lead – This measures the level of support that the IT workforce receives from the company's leadership for cloud migration. It also covers the level of motivation, collaboration, and cross-functionality of the teams. The team structures, executive sponsorship, and budgeting for cloud projects are also covered in this theme.
- Scale – This measures the depth of usage of services that are native to the cloud for reducing overhead in operations, as well as the level of automation you perform for policies and processes which were operated manually. This theme covers how cloud services and resources are provisioned, capacities of workloads distributed, and updates for applications performed.
- Secure – This measures the level of protection in place for your environment from malicious access using a security system that has multiple layers and is identity-centric. This covers the access control systems in place as well as the security strategies and technologies that you use to protect your services.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why it is critical for businesses to adopt new technology

References:
[Services.google.com > The Google Cloud Adoption Framework](#)

Related Material

Question #60 of 60

Question ID: 1495252

Your company uses APIs for providing services to its customers, which span millions across the globe. Now, you want to find a way to specify quotas for your APIs as well as thoroughly monitor their usage. Which Google Service should you use for this?

- A) Firestore

B) Cloud Functions

C) Payment Gateway

D) Cloud Endpoints
- ✖

✖

✖

✔

Explanation

You should use Cloud Endpoints for this. Cloud Endpoints provides a system for managing application programming interfaces (APIs) that enables monitoring, analyzing, and securing APIs as well as specifying quotas for them. You can monitor the key operational metrics for the APIs using Google Cloud Console and gain usage insights with BigQuery, Cloud Trace, and Cloud Logging. You can also utilize Cloud Endpoints Portal to create a website that API users can access to find documentation for the APIs and interact with them. Cloud Endpoints provides API protection using API Keys from Google and JavaScript Object Notation (JSON) Web Tokens. It also has less than 1ms of latency per API call and allows for the deployment of APIs using Google Kubernetes Engine (GKE) and App Engine automatically. Cloud Endpoint utilizes an architecture that is distributed and based on a proxy that uses NGINX ("Engine-Ex"), which is an open source system for load balancing, Hypertext Transfer Protocol (HTTP) caching, and reverse proxy services.

You would not use Payment Gateway for this scenario. With Payment Gateway, you can perform processing of payments that are application programming interface (API)-based. It can scale up to cater to unpredictable surges in usage as well as be extended to third-party insights and anomaly finding tools through APIs. It also allows for payment processing from across multiple regions of Google Cloud's global infrastructure. Payment Gateway also has built-in tools that allow you to monitor, debug, and trace API-based purchases with high availability and less overhead. You can also export data of transactions into a project on Google Cloud using an export API.

You would not use Firestore for this scenario. Firestore is a serverless document database that is scalable and fully managed. It enables fast development for applications for the internet, mobile, and the Internet of Things (IoT), which can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features.

You would not use Cloud Functions for this scenario. Cloud Functions is best suited for when you need to run functional code based on an event or a trigger. Cloud Functions runs your code when events occur from various sources, such as Pub/Sub, Cloud Storage, and Hypertext Transfer Protocol (HTTP). An example is running a process for video transcoding automatically whenever a video is uploaded to Cloud Storage.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Cloud.google.com > Cloud Endpoints documentation](#)

Related Material